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ME-481 & ME-482

Senior Design Project Report

ISO 9001:2015

**FYP/RT-01 Rev # 01 Date: 22
September 2021**

	Name	Signature
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Reviewed by:	Dr. Taqi Ahmad Cheema	
ISO 9001 Section:	Mr. Massab Junaid	
Approved by Dean FME	Dr. Khalid Rehman	
Issued by In-charge student section	Dr. Muhammad Ilyas	

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Design and Development of Vertical Axis Wind Turbine

Senior design project report

BY

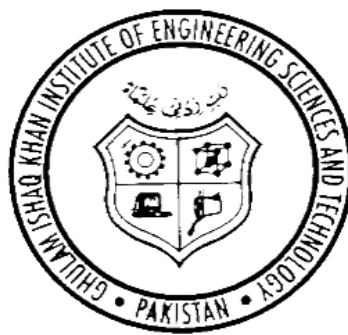
Talha Imran

Muhammad Danish

Muhammad Arbaz-ul-Haq

Supervised by

Dr. Taqi Ahmad Cheema



**Faculty of Mechanical Engineering
GIK Institute of Engineering Sciences & Technology
May 2021**



Design and Development of Vertical Axis Wind Turbine

Senior design project report

By

Talha Imran 2018474

Muhammad Danish 2018253

Muhammad Arbaz-ul-Haq 2018241

Supervised by

Dr. Taqi Ahmad Cheema

submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science

**Faculty of Mechanical Engineering
GIK Institute of Engineering Sciences & Technology**



Faculty of Mechanical Engineering

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8

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This is to certify that senior year design project has satisfied the following:

(i) The design part of the project is completed to a sufficient level. Specifically, the project has achieved.

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(ii) The students have engaged in weekly meetings and demonstrated gradual progression of work, meeting the 6CH requirement.

(iii) The defined Scope / Objectives are.

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a.			
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FYP Mapping with Complex Engineering Problem Attributes

Group No: 8

Title: **Design and Development of Vertical Axis Wind Turbine**

Problem Statement:

Pakistan Govt has seen a rapid growth in number of vehicles. These vehicles became a source for the creation of potential wind energy on highways and motorways. Thus, there is a need to design and install a device which would utilize maximum amount of energy from the wind for the useful purposes.

CEP Attributes Mapping:

(Sr.#1 is mandatory and at least one from the remaining 2~9).

S. No	Attribute	Justification
1	Preamble In-depth engineering knowledge	
2	Range of conflicting requirements	
3	Depth of analysis required	
4	Depth of knowledge required	
5	Familiarity of issues	
6	Extent of applicable codes	
7	Extent of stakeholder involvement and level of conflicting requirement	
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ABSTRACT

This dissertation is the documentation of design, development and testing of vertical axis wind turbine which will be utilized at highway divider to capture the wind energy produced by the moving vehicles and convert it to electricity. Three efficient turbine designs were determined from the literature review after which comparative CFD analysis of these three different vertical axis wind turbines has been done on ANSYS. CFD results shows savonius turbine has low starting torque while H-darrieus is more efficient at high speeds. Torque coefficient for the savonius turbine has the value in range of 0.14 to 0.31, while for H-darrieus turbine 0.06 to 0.45. Keeping highway conditions in mind, a most suitable design in the form of multistage of savonius and H-darrieus turbine, have been introduced. One most important part of multi-stage turbine is rotors, dimensions for which were extracted from the literature review. Shafts were selected on the basis of torsion analysis. Bearing and gears were selected using analytical technique. The design parameters and numerical modeling are also documented. To verify the simulation results experimentally and to further improve the model, a prototype has been fabricated. Prototype has been tested in wind tunnel lab. Power produced by the savonius turbine prototype was in range of 0.319 to 1.87W and 0.267 to 1.84 for H-darrieus turbine while torque coefficient was in the range of 0.12 to 0.28 Nm 0.05 to 0.42 for savonius and H-darrieus prototype respectively. which is up to 92% in conference with simulation results. Fabrication of the full-scale model has been done and testing has been performed by replicating the real-world scenario in the lab using a pedestal fan as a wind source. Turbine rpm produced by the savonius turbine is in the range of 71.67 to 169.27, and H-darrieus has a rpm in the range of 35.65 to 111.18. These results from the lab testing ensures the feasibility of application of the multistage turbine on highways.

Keywords: Wind energy, Highway turbine, wind turbine, stand-alone electrical energy generator, VAWT, Multi-stage wind turbine,

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Nomenclature

VAWT = Vertical Axis Wind turbine

H-Darrieus = H-Shaped VAWT

Helical = Helix shaped VAWT

LTV = Light transport vehicle

HTV = Heavy transport vehicle

T = Torque of Rotor

A_s = Swept Area of Blades

D = Diameter of Blades

ω = Angular Velocity of Rotor

C_T = Torque Coefficient

C_P = Power Coefficient

V = Wind Velocity

U = Rotor Velocity

H = Height of Blades

TSR = Tip Speed Ratio

F_d = Drag Force

C_d = Coefficient of Drag Force

A = Frontal Area (faces the wind)

ρ = Density of Air

P = Power

α = Angle of attack

AR = Aspect Ratio

C_t = Co-efficient of drag in tangential direction

C_n = Co-efficient of drag in normal direction

F_t = Force in tangential direction

F_n = Force in normal direction

Re = Reynold's number

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Chapter 1

INTRODUCTION

This project envisions the design and development of vertical axis wind turbine to produce electricity. It is obvious that worldwide energy crisis is on the horizon. Fossil fuels are becoming rarer. Wind energy is considered as the most appealing alternate for the fossil fuels. It can also be created by the moving vehicles which could be utilized by wind energy converter. The ultimate aim of this project is to design and fabricate a highly efficient turbine. The turbine is ideally designed to be implemented on highways which would extract the wind power from the wind produced by the moving vehicles. The basic idea of designing a highway turbine is to make an effort which would help to overcome the world's energy requirements in a sustainable way.

1.1 Background

Scientific and Industrial technology is advancing towards the promising solution for the growing demand of energy. The key challenge for the modern age is to fulfil the future demand of energy in safe and cost-effective way. For this, numerous research has been done and the new mechanical, technological and material solutions have been proposed, which resulted in a dramatic shift in the development of renewable energy. Renewable energy is the useful energy that is extracted from the renewable resources. Renewable resources (also known as flow resources) are mainly carbon neutral resources which include sunlight, hydropower, wind energy etc. These resources are environment friendly and are easily available with no worries of cost. As the fossil fuels are reducing due to the finite reserves, which results in an alarming condition for the future demand of energy. Moreover, it creates a negative effect on the climate change and causes the environmental implications such as air pollution, global warming. To overcome the traditional method of fossil fuels, scientists are advancing in the field of renewable energy. In the earliest stage, these resources were much expensive, and were not affordable for the common people. But as the global awareness grew, these techniques became more common. Over the last century, wind turbines have emerged as one of the most potential sources of renewable energy and considered as the reliable alternative for the fossil fuels.

1.2 Wind Energy

The atmospheric pressure difference results in the generation of wind. The atmospheric pressure difference occurs due to the sun's even heating of the atmosphere, variations in the surface of earth, and the earth's rotation. This atmospheric pressure difference causes the air to flow from higher pressure to lower pressure end. The atmospheric pressure difference can also occur due to the moving vehicles. As the vehicle moves forward, it compresses front air while pushes the side air, which results in vacuum creation at sides and rear end

of the vehicle. A surge of airflow rushes into the rear and sides of the vehicles to fill this vacuum. Thus, kinetic energy of air is created which can be converted to electrical energy with the help of wind turbines [16].

Wind energy has become one of the most attractive sources of energy as it is scalable and modular. Wind power has been utilized on large and small wind farms and considered as the attractive power source for the distributed power generation applications. Usage of wind energy helps to reduce the dependency on the fossil fuels which is why developed countries have recognized the importance of wind turbines.

1.3 Wind turbine

Wind turbine is a type of turbine which uses kinetic energy of air to rotate the blades of the turbine, in response to which electrical energy is produced. As the wind strikes on the blades of the turbine, the blades start to rotate around the rotor, resulting in spinning of generator through the main shaft which in turn generates the electricity. The amount of energy a turbine can extract from the wind mainly depends on the three factors.

Density of air: The amount of wind power, a turbine can generate is proportional to the density of the air.

Swept area of the turbine: As the capture area increases, the amount of energy in the wind increases.

Wind Speed: The amount of energy a turbine can extract from the wind is directly dependent on the cube of the wind speed. [17]

1.3.1 Types of Wind Turbines

Wind turbines have two main categories: Vertical Axis Wind Turbines (VAWT) and Horizontal Axis Wind Turbine (HAWT).

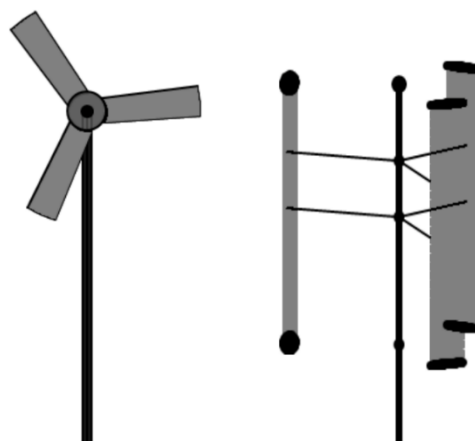


Fig 1-1: a) HAWT b) VAWT [1]

VAWT have axis of rotation perpendicular to the ground while axis of rotation of Horizontal axis wind turbines is parallel to the ground. HAWT works in a stream flow while VAWT works in turbulent flow conditions. Drag force is the major force for VAWT

while HAWT mainly utilizes the lift force. Generally, Vertical Axis Wind Turbines are less efficient as compared to the Horizontal Axis wind Turbine but have more advantages in terms of cost, compact design and are safer than the HAWT.

1.3.2 Vertical Axis Wind Turbines

Vertical axis wind turbine is a type of turbine in which main shaft of the turbine is fixed vertically to the ground. Vertical axis wind turbine did not receive much attention in the past. However, off-grid wind power has gained increasing attention in recent years, owing to the fast advancement of large-scale wind turbines and wind farms. Moreover, Small-scale wind turbines have also been utilized for distributed generation.

1.3.2.1 Types of Vertical Axis Wind Turbines

Vertical axis wind turbines are divided into three types.

- **Darrieus wind turbine:** Darrieus wind turbine has a set of blades attached to the rotating shaft. These blades can withstand at higher speeds due to curvature of the blade
- **Giromill turbine:** Giromill turbine is more like darrieus wind turbine. The H-shape rotor differentiates the Giromill turbine from Darrieus turbine
- **Savonius wind Turbine:** Savonius wind turbine is of the simplest design. It utilizes the drag force, and blades of the turbine cannot rotate faster than wind speed

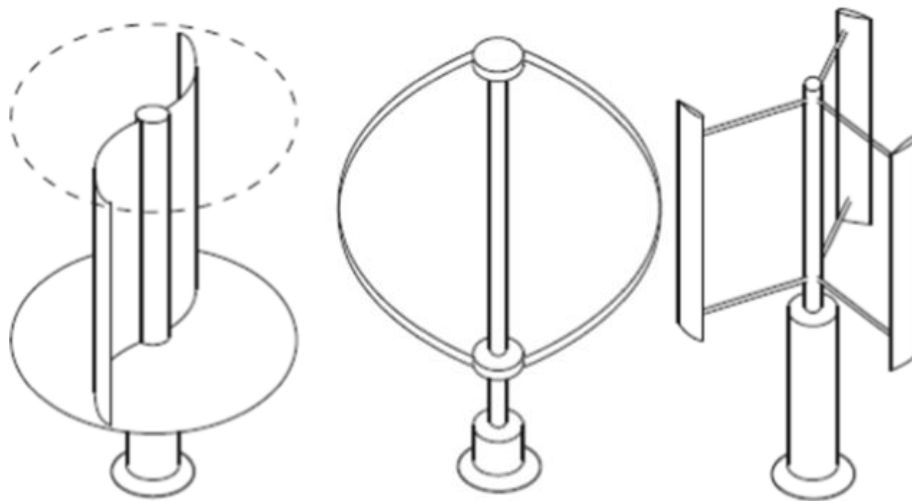


Fig 1-2: a) Savonius turbine b) Darrieus turbine c) H-darrieus turbine [2]

1.3.2.2 Advantages and Disadvantages of VAWT

Advantages

- VAWT does not require alignment (omni directional). There is no need for yaw control
- Maintenance can be done easily for the VAWT as most of its components are located near the ground

- VAWT are quiet and require less space. Thus, they can be utilized in traffic areas (motorways/highways etc.).
- VAWTs can be placed together in a group
- The design and manufacturing process for the VAWTs is much simpler than HAWTs
- It requires much less wind speed to start spinning
- It can operate under the wide range of wind conditions
- VAWTs are Low budget wind turbines compared to HAWTs

Disadvantages

- VAWT generates lower power and are less efficient than HAWTs
- VAWTs are less reliable
- During rotation of vertical blades, bending and twisting can occur which ultimately gives damage to the blades

1.3.3 Highway Turbine

Highway turbines are considered to be the quite revolutionary turbine for the future. When a vehicle moves on the highway, it compresses the frontal air and pushes it to the sides which generates the vacuum at its rear. Bundle of air moves to fill up the vacuum. This motion of air provides a high potential wind energy which could be utilized by converting it to electricity via using wind turbine. A turbine placed on the center of the road is able capture wind energy from both sides of traffic while roadside turbine can capture the energy from one side of the traffic. Thus, turbine is preferred to be placed on the center of the road.

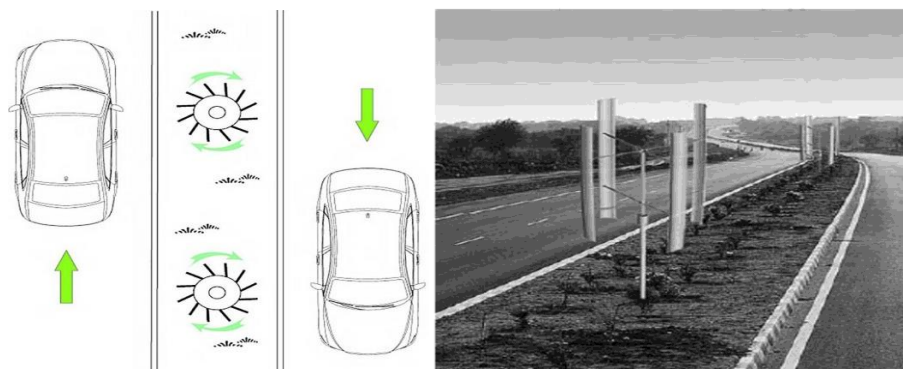


Fig 1-3: a) 2D Visual of Highway Turbine b) Real World Visual of Highway Turbine [3]

1.3.4 Multistage Vertical Axis Wind Turbine for Highways

Multi-staging of the turbine is a modern technique which is getting famous nowadays. In this configuration, multiple rotors of one or more types of vertical axis wind turbine are used. In past simple multi-staged (savonius) turbine has been developed to enhance the efficiency of savonius turbine. To increase the efficiency of simple multistage VAWT, we are introducing the combination of H-Darrieus and helical savonius turbine. This combination has been selected based on simulation results.

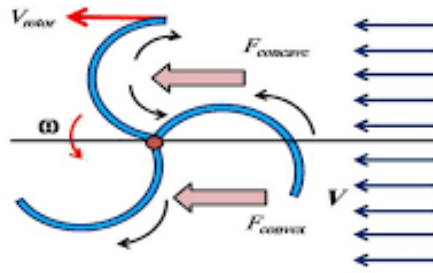


Fig 1-4: Working of Helical Savonius Turbine [4]

Helical savonius turbine is a drag-based turbine which has the working principle more like the simple savonius turbine. The blades of the helical savonius turbine have two faces. 1. Concave (In the direction of wind) 2. Convex (opposite to the direction of wind). When the wind produced by the moving vehicles strikes on the concave side of the turbine, the blades are forced to start moving around central shaft. When air strike on the convex part, air is deflected to sideways due to the shape of convex part. Hence, drag force on the concave side is more than the drag on the convex side. Thus, the turbine starts to spin due to the differential drag force.

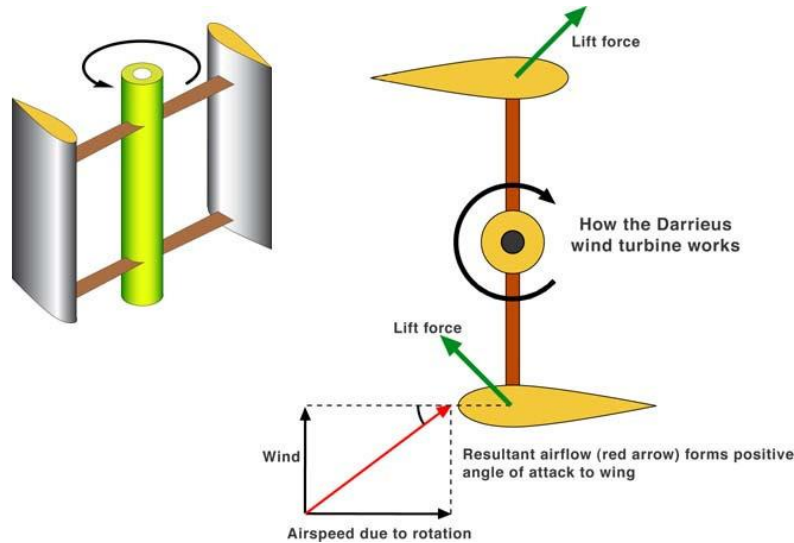


Fig 1-1: H-darrieus Working Principle [5]

H-Darrieus turbine utilizes the lift force for the motion. When air moves around the structure of the turbine, a suction has been produced on the front side of the blades. Thus, the blades of the turbine start rotating as shown in the fig. 1-4. H-Darrieus turbine experiences less drag force, as compared to helical Savonius turbine, due to the shape of the blades. It has the ability to rotate faster than the speed of air, once it gets going.

1.4 Motivation

- The initial motivation for this project was the school bus accident. A school bus was overturned on Gojra road due to lack of visibility at night. This school bus was

carrying 27 students and 3 teachers. After reading this news, we thought to improve the lighting system at highways. After doing lots of research, the best way to improve the lighting system has been found that is the installation of turbine on highway which would utilize the wind energy produced by the moving vehicles [18]

- Government of Pakistan manages more than 0.85 million vehicles per day on motorways and highways. One can imagine how much wind energy goes useless. Currently, there are no plans to utilize this energy. Installation of wind turbine, which would extract maximum amount energy of from the wind, would be much helpful to overcome the energy crisis in Pakistan [19]

1.5 Problem statement

Pakistani government has seen a rapid growth in number of vehicles. These vehicles became a source for the creation of potential wind energy on highways and motorways. Thus, there is a need to design and install a device which would utilize maximum amount of energy from the wind for the useful purposes.

1.6 Aim and objectives

The aim is to design and fabricate a Vertical Axis Wind Turbine which would be used on highways to generate energy from passing vehicles. The following objectives for the Senior Design Project have been set to achieve the above-mentioned aim.

- Assessment of Savonius turbine
- Comparison between different types of VAWTs
- Design proposal
- Design analysis and Calculations
- Simulations
- Sustainability Analysis
- Assembly and Fabrication
- Testing of new model

1.7 Scope of Product

- Wind energy convertor
- Multistage turbine
- Three blade Savonius turbine
- Three blade H-Darrieus turbine
- Applicable on highways

1.8 Scope of the Project

This project is divided into three main sections.

- Design of the highway turbine
- Fabrication of the highway turbine

- Testing and analyzing the turbine

All three group members will be contributing to the each of these three sections. The design section will detail the specifications of the turbine. Fabrication section will focus on the development of multi-stage Vertical Axis wind turbine. Third section will be related to the assessment and analysis of the turbine.

1.9 Tentative Project Timeline

Table 1-1: Project timeline

Tasks	July-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22	Apr-22	May-22
Previous model testing											
Literature Review											
Model Designing											
Simulation											
Presentation											
Procurement for prototype											
Fabrication and testing of prototype											
Procurement for full scale model											
Fabrication of full scale model											
Testing											
Report Completion											
Open House											
Report Submission											

1.10 Report Outline

Chapter 1 introduces the project, background, motivation, problem statement, aim and objectives, scope of work and project timeline.

Chapter 2 explains the literature review and discusses the research gap found in literature survey.

Chapter 3, Design and analysis: explains the design methodology, process flowchart, governing equation, mathematical modeling, codes and standards which are utilized in the project.

Chapter 2

LITERATURE REVIEW

Energy is one of the important sources to carry on any task. Many of them are non-renewable such as gas, coal, and liquids fuels/oil. When they burnt the amount of carbon dioxide produced by the fossil fuels are threatening to the environment. Other forms of obtaining the energy to obtain the work are sustainable and clean energy. Wind, tidal, biogas and solar energy are used nowadays to generate the energy to fulfil the requirements. Wind energy is the purest form of clean energy which is used to generate electricity. It is also a cheap and less toxic than all other sources of energy.

Wind was firstly used for many purposes, for example navigation and agricultural usage for thousands of years ago. People used the wind to sail their ships in the river. People in Persia and China were pumping water and grinding grain using wooden wind-powered water mills by 200 BC. These tiny wind turbines were primarily utilised in rural regions and for minor jobs. The concept of wind turbine originated with the invention of airplane and the scientist seems the potential in wind to do some useful work, and then the first windmill was invented by Charles Bush in 1888. But wind turbine is commercialized in late 1990's.[20]

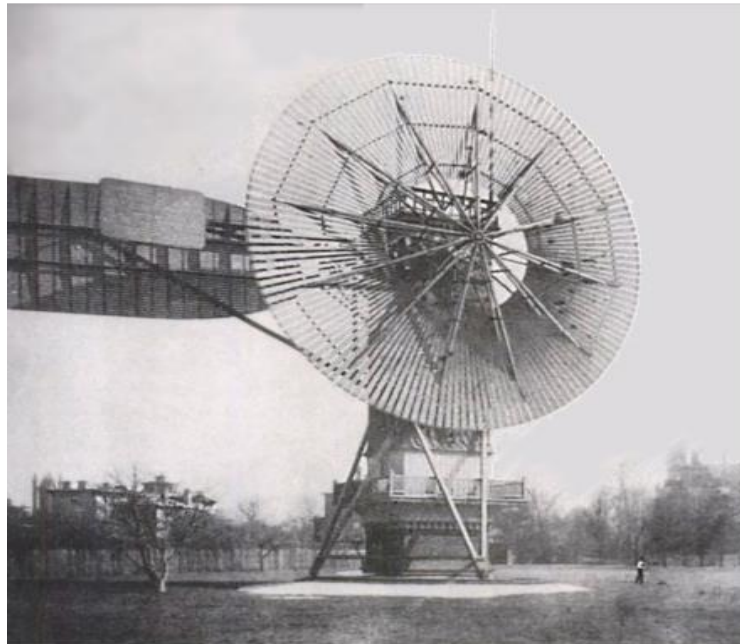


Fig 2-1: First Windmill by Charles Bush[6]

One of first offshore wind turbine area was built in Denmark,1911 and has the capacity to produce 495MW. Offshore wind turbines are preferable in the past as in the seas and oceans there are higher windspeed and hence capable of producing more power. Walney wind farm is one of the largest offshore wind turbines. On the other hand, onshore wind

turbines are easier to construct and hence the cheaper way of power generation. It costs half as much to produce power per kWh as offshore wind energy and a quarter as much as solar panels.[21]

HAWT and VAWT are two main types of wind turbines. HAWT are wind turbines with a center of rotation that its axis of rotation is parallel to the ground. A rotor shaft and an electrical generator are located at the top of the tower, back to blades of HAWT and blades must face the wind. Due to turbulence, the turbine is generally positioned upwind of its supporting tower to generate more power. Turbine blades are typically angled forward to face the wind.[22]

The center of rotation of a vertical axis wind turbine is perpendicular to the axis of rotation and the rotor are located near the ground. Most common designs from the past which the manufacturers prefer are Savonius, Darrieus, Helical and Giro mill. Each of the design has its own disadvantages and advantages with the significant variations in design and large bending moment in the rotor blades.

Now a days scientist consider wind to meet the future demands of energy. But the main issue with extracting energy from wind is that wind speed varies with each hour, and it is impossible to meet the requirements of stabilize power, therefore wind turbine is always use in conjunction with the batteries to stabilize the input.

The power generation by vertical axis wind turbine (VAWT) was investigated by Niranjana. S.J. In this article, power is created by mounting a wind turbine on a highway which used the drag of moving vehicles to rotate the rotor of VAWT. The rotor of the windmill rotates and creates power when a vehicle passes over the road at a high speed. According to research, a VAWT is independent of direction and captures the wind from any side, and generates 1 KW (Kilowatt) of electricity at a vehicle speed of 25m/s. The efficiency of a VAWT (vertical axis wind turbine) may be improved by changing the blade shape, diameter, length, and its form.[23]

Further design considerations and construction of a VAWT were studied by Roy et al. VAWT is one of the most significant forms of wind turbines which generates electrical energy. The wind turbine's primary rotor shaft is linked vertically to the dynamo (which converts the mechanical to electrical and acts as the generator) and gearbox, which can be located close to the ground level. Performance parameters such as power output versus wind speed or angular velocity must be modified to achieve optimal efficiency in order to compete with other energy sources and make the process economically and ecologically acceptable. The results of the experiments indicate that installing a wind turbine on top of a building in an ideal position or between the midpoints of roads for electricity generation is a smart idea.[22]

Because of the methodologies utilized for design, energy extraction from wind is one of the emerging technologies. Today's research approaches are generating more efficient turbine blades. Over the last several years, the yearly energy production of turbines has grown dramatically, while the weights of turbines and the noise they generate have decreased significantly. In 2016, an experiment was conducted on the design and analysis of VAWT. The purpose of the research was to determine the blade's structural response to

the incoming air and the pressure differential between the concave and convex surfaces of the rotor blade. Some of the VAWT's benefits over the HAWT (Horizontal axis wind turbine) were also highlighted in the article. VAWT is unaffected by wind direction and does not require any additional force to operate. It has a strong starting torque, is easy to construct, maintain, and emits little noise and sound than HAWT. On two blades Savonius rotor, the findings clearly showed that the concave blade section had high pressure while the convex portion had low pressure.[24]

The study was done on two and three blades of Savonius turbine and concluded that which one is suitable and better for the extraction of power from wind. The experimental analysis was done in the wind tunnel and the dimensionless parameter like torque coefficient and the power coefficient was evaluated as a function of tip speed ratio. The experiment was concluded that three rotor blades have more moment of inertia than the two rotor blades and the two rotor blades have high initial torque than the other. The experiment was done on no load conditions to achieve the maximum acceleration at low wind speed.[25]

Nikam et al. conducted a study of the literature on vertical axis wind turbine blade design of rotor and development. Two types of wind turbine, such as horizontal and vertical wind turbine, is widely utilised for power generation, according to this article. The horizontal windmill is often utilised in large-scale applications that necessitate a lot of area and a lot of money. VAWT on the other hand, is ideal for domestic use. The shape and orientation of the blades of a wind turbine have an impact on the generation of energy as it greatly. Set the correct parameter for the blade design to optimize this. The testing results show that the turbine's blade dimensions, and its structure plays a vital role in its performance and power output. The optimized blade parameter and its specification can help with power generation.[26]

Hossain et al. looked into the construction and design considerations of a 1/3-scale vertical axis wind turbine for the electricity generation. Belt power transmission system is used to generate energy from a wind mill rather than using gear train. Different wind speeds are used in the experiment to analyse the best design, and the power generated by the windmill is determined. The testing results show that at a wind speed of 20 m/s, 567 Watt of power is generated, while at a speed of 25 m/s, 709 Watt of power is generated. As a result, when the velocity is high, the power generation increases.[27]

The combination of both Darrieus and Savonius design was studied, and it is clear that the efficiency of the combined VAWT is better than the individual one. Moreover, the efficiency mainly depends on the climatic conditions, aspect ratio, tip speed ratio and other geometry parameters. In this article it is clear that lower aspect ratio increases the power generation by the windmill. A multistage generator is a two-stage generator with the same rotor size. The multistage design has the following characteristics discussed in the other article.[28]

- This setup is capable of meeting at least 10% of the more consumption power than the individual installation of windmill
- When the wind speed is low, the gear configuration might improve the number of rpm

- This turbine is best suited for heights of 8 to 10 meters above ground level. Because of the velocity at ground level
- It is possible to enhance output by combining an alternator with a gear system, but this would raise the machine's cost needlessly
- Reinforced glass fibre is an alternative to aluminum alloy for turbine blade material since it is more elastic. However, it is more expensive but gives high initial torque

Angle of attack is one the main key point that effects the efficiency of VAWT. It was studied by Akhil et al. The aerodynamics shape of the rotor combined with the optimum angle of attack gives the peak value of power extracted from the wind. The blades were designed using Q-blade software and the power extracted is determined by the blade elemental theory.[29]

2.1 Research Gap in Literature Review

As concluded in the previous discussion, various ideas have been offered for obtaining power from the wind, but no combination can be used. If it is employed, both turbines are mounted on the same principal shaft, reducing the efficiency of the other turbine, which is typically used at lower speeds. When both turbine models are mounted on the same shaft and one has a higher efficiency than the other, the efficient turbine suffers from inertia and frictional forces, causing the entire system to suffer.

Due to the fact that both turbines are positioned on the same principal shaft, only one dynamo (generator) is utilised to convert mechanical energy to electrical energy, which is also a disadvantage of the system. For the turbines, the generator frictional forces and back emf are stronger in a single generator. For a multiturbine on the same shaft, the overall losses are not as effective.

If both turbines are positioned on separate shafts, these types of losses may be reduced, and this approach also minimize the shaft weight and its working by mounting two turbines on distinct shafts, and because the output from both generators is loaded, the back emf in each generator is significantly lower than if both turbines were mounted on the same shaft.

The turbines are installed on two different shafts, thus the power extraction from one shaft is not greatly influenced by the power extraction from the other shaft, and the single entity may extract more power from the vehicle's incoming wind. When a light transport vehicle (LTV) passes by, it affects the lower part of the turbine more than the entire VAWT, resulting in power being extracted from only one turbine, whereas when a high transport vehicle (HTV) passes by, it affects both parts of the turbine, resulting in power being extracted from only one whole turbine. The extraction process will be considerably smoother and more efficient as a result of this. We may sum up our findings from this project's research as follows:

- Each turbine in a multistage system has its own generator
- Different turbines are mounted on different shafts
- On the lower side, a turbine that can function at low speeds and has lower initial torque is fitted

- On the plus side, choosing a turbine with a higher beginning torque and a higher efficiency

2.2 Summary of literature Review

A study was compiled from several research publications, and experimental and analytical data were analysed in order to examine the VAWT's areas of continuing growth. It has been suggested that combining two turbines is the most efficient way to harvest wind energy. Furthermore, if both are mounted on different principal shafts, they will produce greater power than if they are put on the same principal shaft.

Chapter 3

DESIGN AND ANALYSIS

3.1 Design Selection

The idea of vertical axis wind turbine on highways to extract wind energy produced by the moving vehicles is not entirely unique. Many attempts have been done in this regard, but none of them has been able to propose a design which would fulfil the requirements. To find the best possible design for highway turbine, it has been decided to take the help from the previous research. From the literature survey, discovered three most appealing designs alternative which could be utilized for this purpose. These three design alternatives are written below.

- Savonius Turbine
- Helical savonius turbine
- H-Darrieus turbine

Each design has its own pros and cons. A brief discussion has been done on every design alternative. Furthermore, simulation have been done on the ANSYS (software). The SST $k-\omega$ (this model gives the accurate value near the surfaces of the VAWT on which the wind strikes, and SST is selected because standard $K-\omega$ model doesn't have the capability to switch between two models when needed like $K-\omega$ and $K-\epsilon$ model. The fluid in the consideration was air as it changes the direction when strikes with the VAWT and it is the force that is solely responsible for the rotation of rotor. The meshing with 0.05m size has been utilized. The results of the simulation are compared to find the most suitable design for the highway turbine.

3.1.1 Savonius Turbine

VAWT are gaining more and more popularity in the industry of renewable energy. The most common design of the vertical axis wind turbine is of the Savonius turbine. The Savonius turbine is a drag type device, invented by the J. Savonius in 1922. The Turbine was consisting of two half-cylindrical blades. These blades were arranged to form S shape as shown in the figure (3-1). However, Savonius turbine with more than two blades were also built. When air strikes on the concave part (advancing blade) of the blade, it forces the blade to spin around its shaft. Whereas, when air strikes on the convex part (returning blade) of the blade, it deflects the blades towards sideways around the shaft. The concave side of the blade experiences more drag force than concave side of the blade, as drag coefficient for the convex half is less than the drag coefficient for the concave half. This difference of drag, forces the rotor to spin and develop mechanical power. As the wind which strikes on the convex half of the blade opposes the rotational motion produced by concave half. Thus, much of the wind power goes useless. For this reason, Savonius turbines are considered to be less efficient.

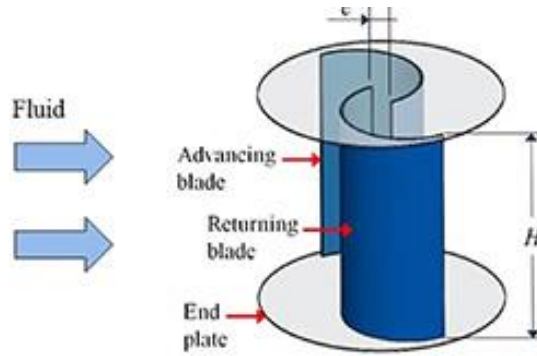


Fig 3-1: Savonius Turbine [7]

Savonius turbine is also well known for its simple design structure. Blades of the turbine are of cylindrical shaped, so there is no need for technical specialist to fabricate the savonius turbine. The fabrication process can be done even in small workshops. Due to their simple design, they are more common in developing countries. Rotor of the savonius turbine runs at low speed and has the high solidity.

Savonius turbine can be improved by the addition of endplates. Addition of end plates to the savonius turbine increases the efficiency up to 36% as compared to the simple savonius turbine [12]. The increase in efficiency is due to the blockage provided by the endplates to airflow. Thus, the air could not escape from upper and lower side which provides the higher fluid flow momentum to blades, which ultimately increases the positive torque. Moreover, the addition of end plates enhances the power coefficient as it prevents the air from exiting which results in the increase in pressure difference between concave and convex side of the blades.

Even though, there are many methods proposed to improve the efficiency of a turbine to a higher level, but still, it remains less than the other types of wind turbine. Rotor of the savonius turbine have low power coefficient as it works on the low tip speed ratios.

3.1.1.1 Simulation

The simulations of different designs were done on the ANSYS FLUENT. It had been decided that the wind with 5 to 10m/s would be utilized in the enclosure. The speed of the air was deducted from the literature survey, according to which when a car moves with a velocity of 80 to 100km/h, wind is produced with the velocity given below. The VAWT was placed in the enclosure so that it would show more realistic results as in the real-world scenario i.e, when VAWT was placed on the road, traffic was moving on both sides due to which air strikes on both sides of the turbine. Boundary conditions was given as follows

- Velocity at the inlet of the enclosure was given to be 5 m/s to 10 m/s
- Atmospheric pressure at the outlet of the enclosure was given
- All the other sides are walls (sides, top and bottom of the enclosure) were given with the no slip boundary condition

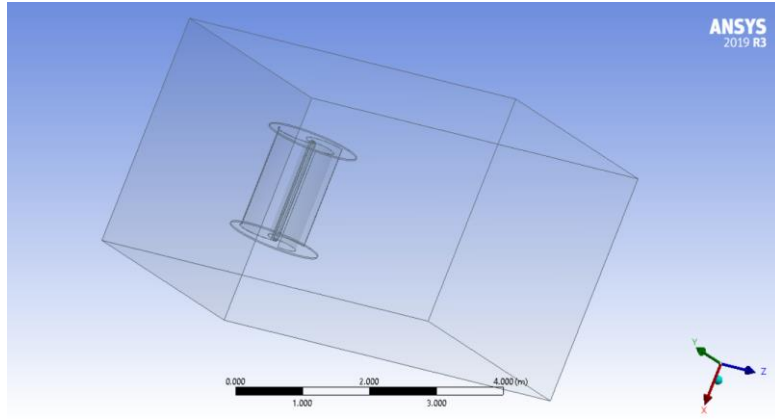


Fig 3-2: Savonius Turbine in enclosure

The coupled method, also known as the pressure-based coupled algorithm, was utilized in the solution analysis because it allows for full pressure-velocity coupling. Hybrid initialization technique was used as it is based on solving of Laplace's equation to determine the pressure and velocity parameters and the solution of the equation converges to 1×10^{-7} . Five hundred iterations were carried out for the solution to converge.

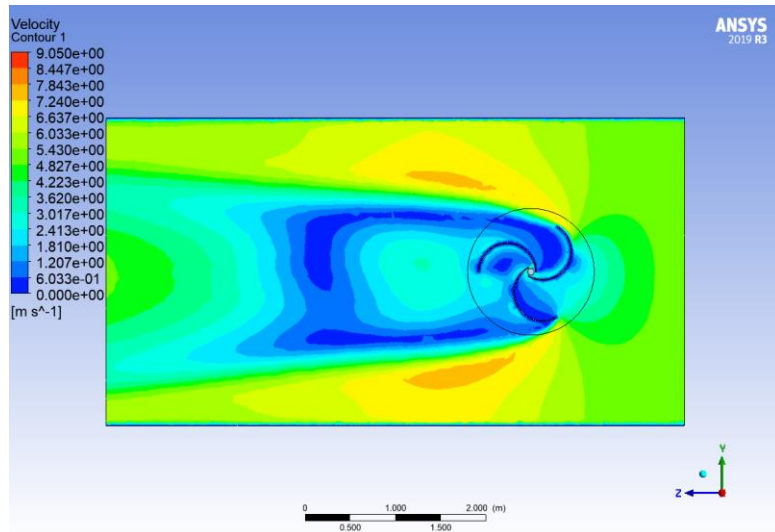


Fig 3-3: Velocity contour

Figure 3-3 shows the variation of velocity throughout the enclosure. Wind is flowing from the inlet given in the right side of the figure 3-3. As the wind flows and interact with the VAWT it will rotate as the velocity on the one of the sides has the greater effect on the rotor than the other side and hence the greater drag force will be obtained by the square law formula.

$$F_D = \frac{1}{2} C_D A \rho v^2 \quad (3.1)$$

Where, F_D is the drag force, C_D is the coefficient of drag force, A is the frontal area (faces the wind) and V is the wind velocity

Figure 3-4) shows the pressure variation in the enclosure. Pressure at concave side is greater than the convex side, so drag will be greatly acting on the concave side.

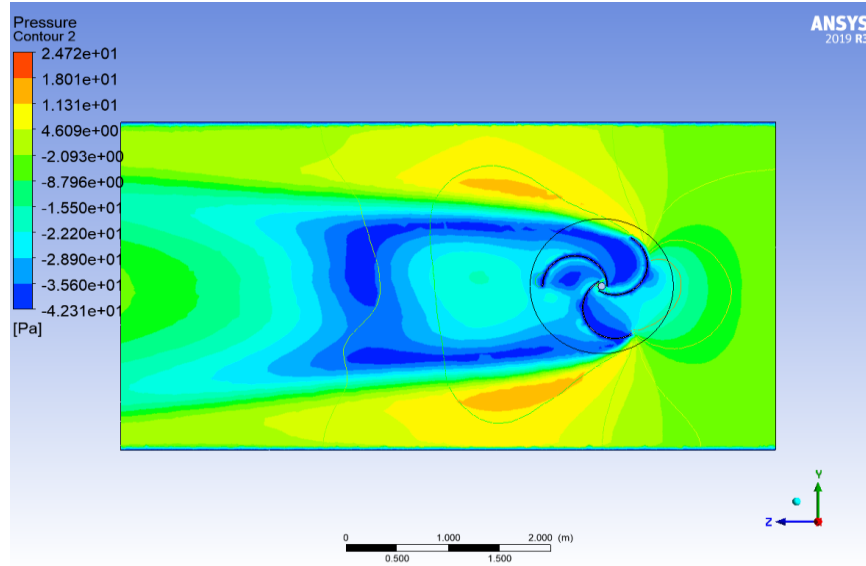


Fig 3-4: Pressure contour

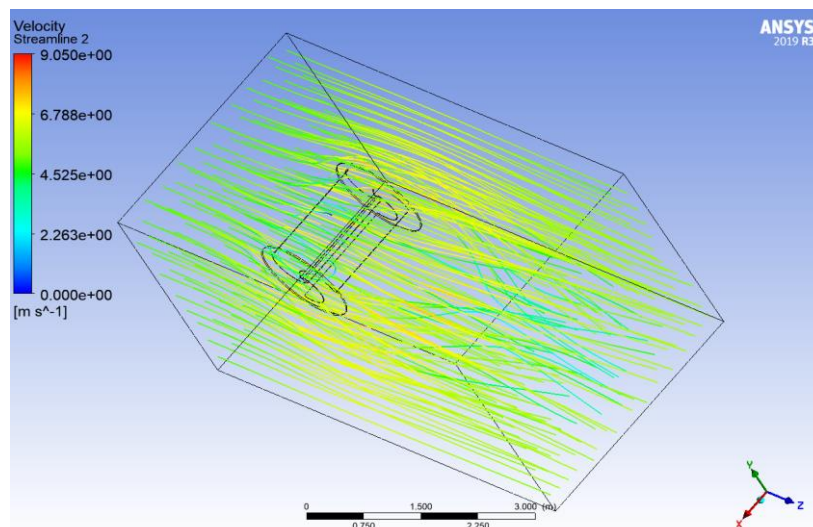


Fig 3-5: Velocity streamlines for 3D Simulation

3.1.2 Helical Savonius Turbine

Vertical axis wind turbines are gaining more and more popularity in the industry of renewable energy. To improve the efficiency, lots of design for vertical axis wind turbines have been proposed. One of these innovative turbine designs is the helical vertical axis wind turbine. The helix savonius turbine is the latest version of savonius vertical axis wind turbine. The main difference between helix savonius turbine and typical savonius turbine is the design of the blades. The helix savonius turbine has the helical blades which resemble

to a DNA strand. Helical vertical axis wind turbine is considered as the more efficient wind turbine because, rather than having simple straight blades, it has the curved blades twisted helically around the axis of the rotor.

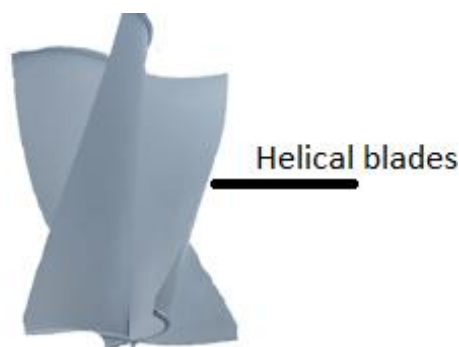


Fig 3-6: Helical savonius turbine blades

The main advantage of the helix savonius turbine is the lower threshold wind speed. As the typical savonius turbines require a higher wind power to get blades spinning while helical blades of the turbine require much less wind speed to start spinning. Thus, helical vertical axis wind turbine starts producing electricity even in the presence of obvious low wind. Once the motion of the blades started, helical blades hardly need any air to continue the rotation. Moreover, Helical turbines are considered to be safer at higher wind speed as compared to other types of turbines. As most turbines has a speed limit, after which they have to be locked down. Thus, Helix Savonius turbines are more suitable for the extreme conditions.

Furthermore, the regular turbines typically have extremely high speeds at the blade tips. Anyone in the nearby will be annoyed and distracted by the high-pitched screaming sound produced by the high-speed wind. The design of helix savonius turbines allow for slower speed over the full length of the blade due to which most of the irritating noise associated with wind turbines is eliminated. Thus, helical vertical axis wind turbines are quieter than the regular turbines. Besides that, helical wind turbines also share some advantages of other types of vertical axis wind turbines. Like the typical savonius turbines, bird problems have rarely been an issue with a helical wind turbine. Helical savonius turbines can also be placed near the ground. For this reason, these types of turbines are more common in the residential areas [30].

By considering all the perk and benefits offered by the helical vertical axis wind turbine, it has been decided to consider the helix savonius wind turbine as the first design iteration to improve the efficiency of the wind turbine.

3.1.2.1 Simulation

The CFD analysis of helical savonius turbine was carried out on ANSYS FLUENT. The wind velocity of 5 to 10 m/s was used in the enclosure in which the VAWT is placed. This depicts the real-world scenario as when the VAWT is placed on the road the air gets hit by both of its sides. This CFD simulation was based on current literature, which says that if a car passes by at 80 to 120 km/h, a similar wind velocity will be produced that effects the rotor of VAWT in the same fashion as in the case of ANSYS FLUENT. The prior design

simulation's boundary conditions were maintained which were inlet velocity in the range of 5 to 10m/s, atmospheric outlet pressure and no slip boundary condition.

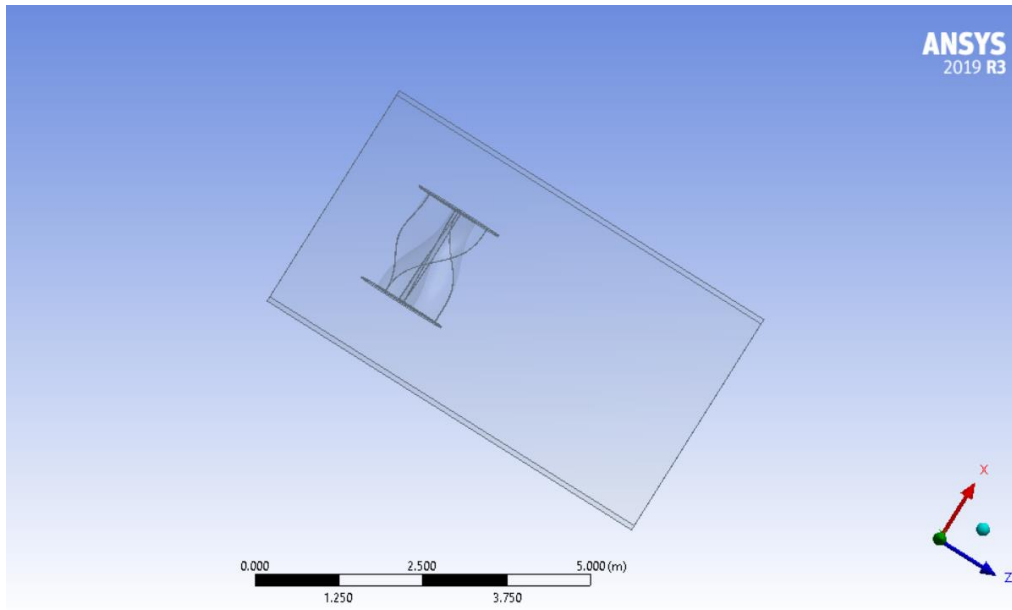


Fig 3-7: Helical savonius turbine in enclosure

For the solution analysis, couple method was used due to the same reason discussed in the previous design simulation. The pressure and velocity parameters were determined using the hybrid initialization approach, which is based on solving Laplace's equation, which converges to 1×10^{-7} . For the solution to converge, five hundred (500) iterations were performed.

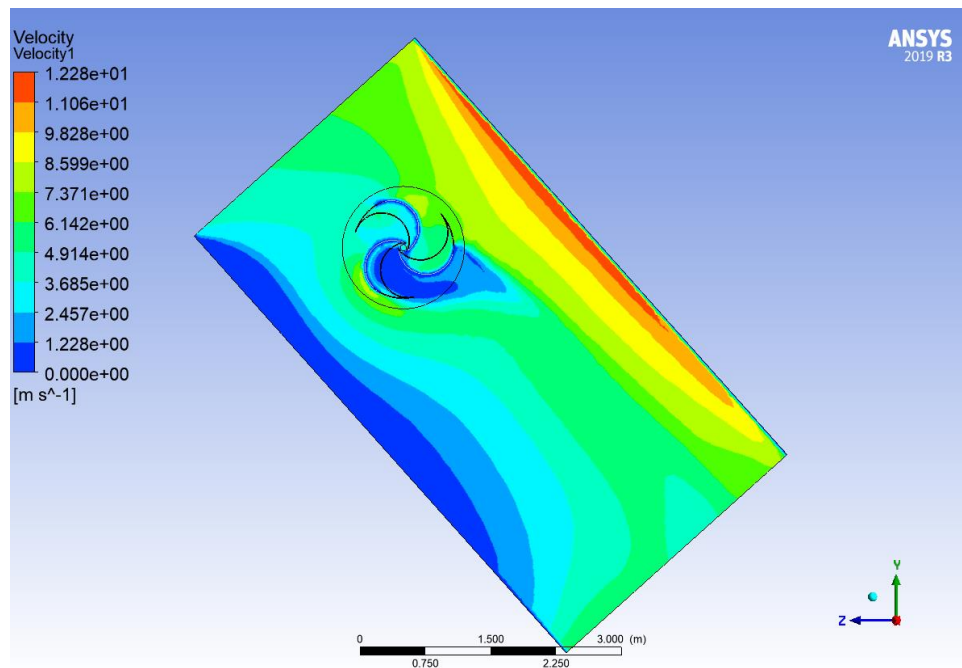


Fig 3-8: Velocity contour

The velocity variation in the enclosure is shown in the fig. 3-8. The maximum velocity obtained in this case is 12.28 m/s. The wind is flowing left to right in the enclosure. Due to increased drag and lower pressure on the remainder of the area, the velocity contours indicate a significant speed build up on the concave side of the advancing blade. The air slides down the concave side of the moving blade, forcing it to rotate, as seen by the velocity contours. The air then passes through the overlap zone, where the air accelerates owing to a constriction created between the rotor blade edge and the center shaft, causing a shift in momentum, and finally creating higher force. This validates the decision to include overlap in the turbine's design.

Figure 3-9 shown the pressure variations and in the figures 3-10 shown the velocity streamlines and it is clearly shown that one side of the helical savonius bears the high-pressure region and the other side has the low-pressure region thus, due to the pressure difference in the front and back side of the helical savonius surface, it rotates. In the velocity streamlines figure, it is shown that the particles striking the surface of the blade will be the reason for the drag on the surface and due to the forces of drag it moves in the direction of the air from either side. So, it is clearly concluded that the helical savonius is based on the drag concept and it rotates die to the drag forces acting on the surface of the blades.

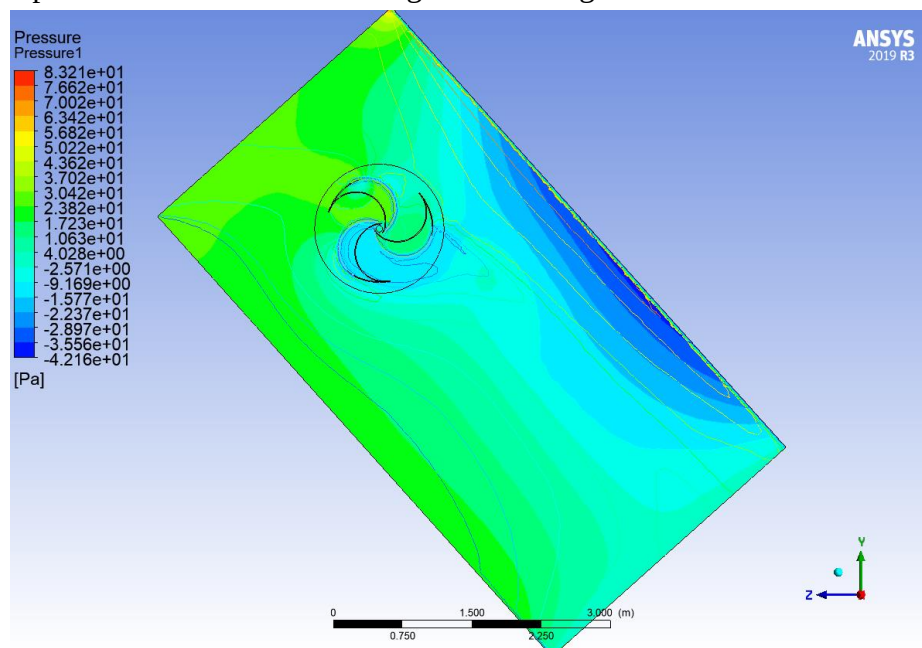


Fig 3-9: Pressure Contour

The pressure contours also suggest the same findings as obtained from the velocity contours. Pressure at the convex side of helical VAWT is lower than the one of the concave sides and hence greater drag will be obtained in that case on the concave front side.

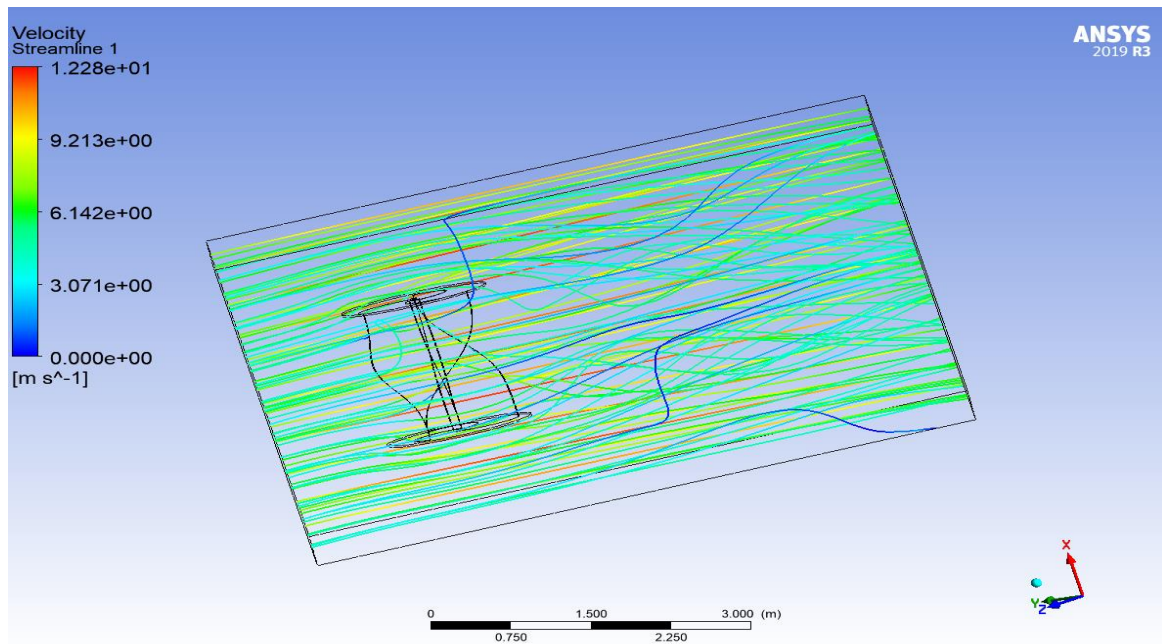


Fig 3-10: Velocity Streamlines for 3D Animation

3.1.3 H-Darrieus Wind Turbine

Darrieus turbine is VAWT, invented by the George Darrieus in 1926. Originally the design had number of curved airfoil blades which were mounted on the shaft. It was symmetrical with zero rigging angles and required an electric support to start. Later, different designs of blades had been introduced to improve the different parameters of Darrieus turbine. One of the most common types of the design was H-bar design.

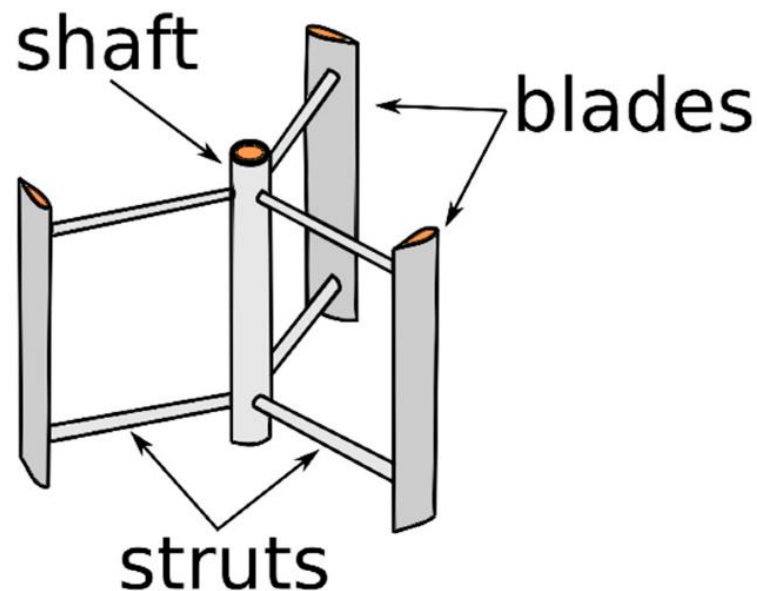


Fig 3-11: H-Darrieus Turbine

H-Darrieus turbine is the VAWT which utilizes the lift generated by the vertical air foils. It is a type of Darrieus turbine in which straight vertical blades were used instead of long “eggbeater” blades. These blades were attached to main shaft through the horizontal

supports. When air flows around the structure, suction on the front side is created due to which blades start moving. Due to the shape of the blades, H-darrieus turbine experiences much less drag force than the savonius turbines. The blades of the H-Darrieus turbines have the ability to move faster than the wind speed once it gets going.

Torque varies significantly in H-Darrieus turbine. This variation can be decreased by the number blades. The higher the number of blades used, reduces the variation in torque. Three bladed turbines are considered to be the most appropriate choice. They also offer the self-starting ability. Torque pulsation can also be reduced by varying the pitch along the blade. Moreover, power coefficient for Darrieus turbine is higher as compared to Savonius turbine on high-speed winds.

H-Darrieus turbines are cheaper as compared to simple Darrieus turbine. They are easier to build. H-Darrieus turbine can be effective in turbulent wind conditions. however, sometimes it is difficult to maintain the steady state of rotation.

The angle of attack on the blade and azimuth angle constantly varies which makes it aerodynamically more complicated. The key challenge occurs in the design phase of the blades. As it experiences high bending moment. The reason behind the high bending moment is centrifugal acceleration. Thus, the designing process becomes more challenging. Surface roughness is also considered to be one of the key factors which affects the performance of the turbine. It is expected that surface roughness increases as the time passes due to erosion or contamination. According to Priegue and Stoesser, it reduces the turbine performance. This effect become much more at higher Reynold number. Reynolds number varies for H-Darrieus turbine even at steady tip speed ratio. It is due to dependence on relative velocity which changes due to the variation in wind velocity. It is also dependent on the blade cord length. Reynolds number does not affect the performance of the turbine but also predicts lifespan of the machine [31].

3.1.4 Simulation

The CFD analysis of H-Darrieus turbine was performed on the ANSYS FLUENT. The boundary conditions were kept same as of the previous two designs. Couple method (also used in the previous designs) was used for the solution analysis. Pressure and velocity were determined via using same technique as the previous. and to converge the solution 500 iteration had been performed. The turbine in the enclosure is shown in the figure 3-12.

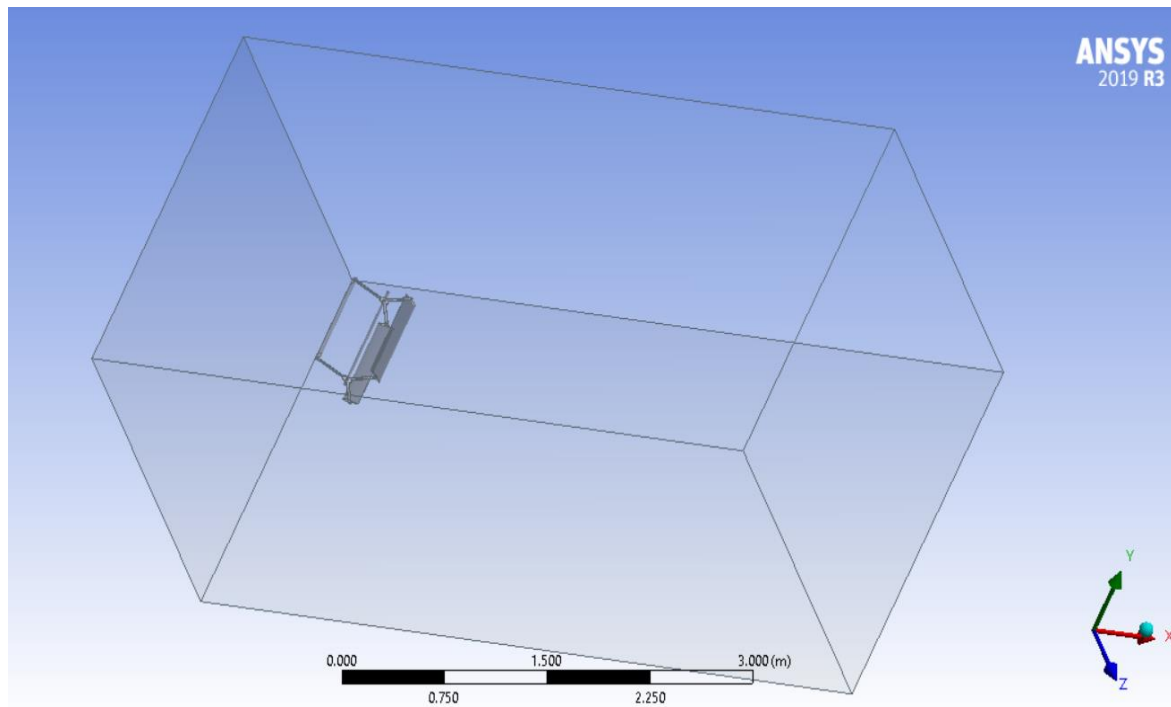


Fig 3-12: H-Darrieus Turbine in enclosure

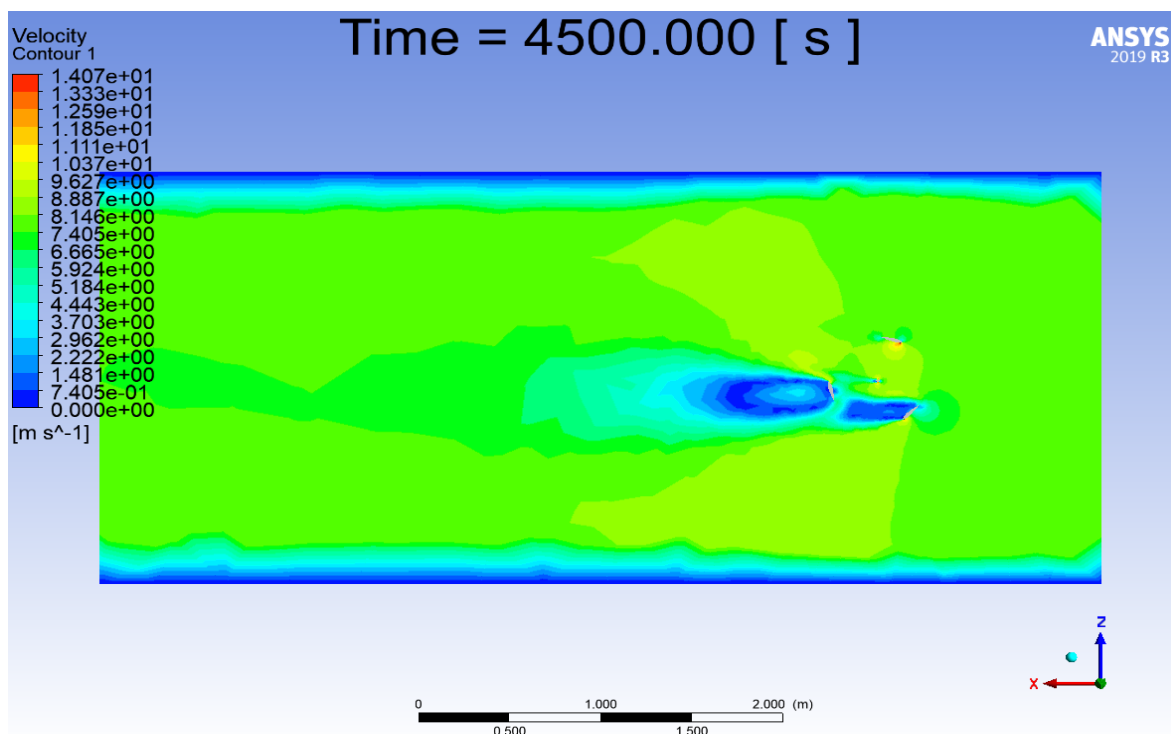


Fig 3-13: Velocity contour

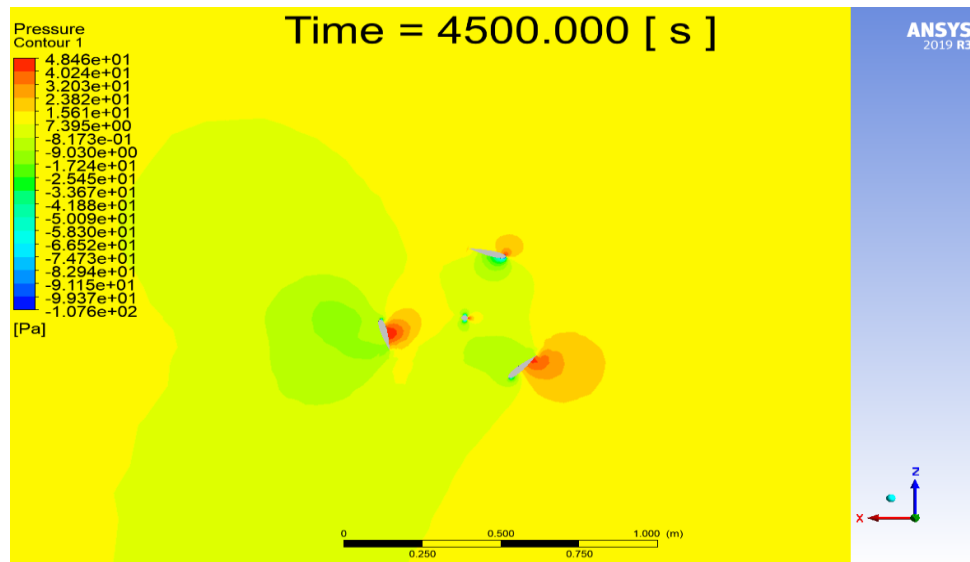


Fig 3-14: Pressure contour

3.1.5 Final design

A comparative CFD analysis of three different types of turbines has been performed on ANSYS. From the results, it is clear that helical savonius turbine is better for low wind speed while H-darrieus is more efficient for the higher speeds. By analyzing our requirements and conditions, we are introducing the multi-stagging of helical with H-darrieus turbine to get the maximum efficiency at any speed. Helical savonius turbine is placed on the bottom part while H-darrieus turbine is used on upper half.

3.2 Design Methodology

The complete design of VAWT was divided in to three main parts.

- Designs of helical blades
- Designs of H-Darrieus blades
- Design of base

Design of H-Darrieus was divided into further parts

- Aerofoils
- Struts
- Retainers

Design of helical savonius were divided into further parts

- Rotor blades
- End plates

Parts that were designed for both of the models are:

- Rotor blades
- Impeller Shaft and plugs
- Bearings
- Base and top frame

3.2.1 Description and Justification of Design Method

Design of the rotor plays a crucial role as the efficiency of the VAWT is highly dependent on it. To find the design having high efficiency and self-starting ability to use on highways, three most appealing designs have been discovered from the literature survey. These selected designs were CAD modeled after which simulation has been done on ANSYS Fluent. By analyzing simulation results, most suitable design has been proposed which is multi-stagging of H-darrieus and helical savonius turbine (with end plates). Dimensions of the rotors have been decided after studying literature review thoroughly. Shaft diameter has been calculated such that it can bear the weight of blades and have the ability to stand in stormy conditions. Bearings and gears were designed using the diameter of the shaft. Material and dimensions for the base and top were selected while keeping width and weight of the assembly in mind. After all the mechanical components have been designed, electronic components were selected.

The design process has been started with the design of the blades as shaft calculations are dependent on the size and weight of the blades. Once the blades and shafts were finalized, bearings and gears were designed using shafts and blades dimensions. Base structure and supporting rod were also selected using finalized parameters of blades and shafts. 3D model has been updated and to fabricate the turbine 2D drawings for each and all components are generated.

The following figure shows the proposed initial design.

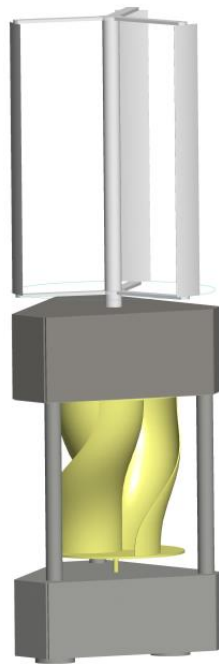


Fig 3-15: Initial proposed Design

This design consists helical rotor at bottom half while H-darrieus is used at the upper half of multi-stagging turbine. Each rotor has its own shaft and base. Gears, bearings and

electronic components are installed at each base. Three supporting rods in the lower half are used to carry upper base and H-darrieus rotor and to stronger the structure.

3.3 Design Process Flowchart

The process flow chart shows the sequential steps of the design process. Starting from the literature review and choosing the best design through CFD analysis to fabricate the final design by doing the mathematical modeling, design of mechanical parts and the selection of electronic component.

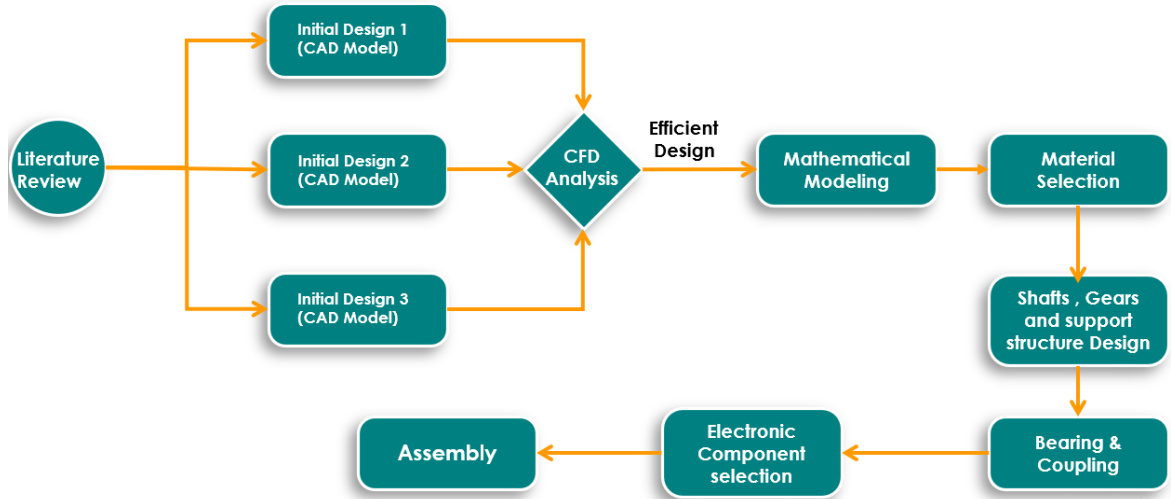


Fig 3-16: Flow Chart

Multi-staging is of two types

- Common generator multi-staging
- Separate generator multi-staging

Separate generator multi-staging is more efficient than the common generator multi-staging. Thus, our design consists of two separate generators with two separate shafts for each of rotor.

3.4 Governing Equations

3.4.1 Helical Savonius Turbine

Formula given below is to calculate the torque [32] produced by turbine after the striking of wind.

$$T = \frac{1}{4} \rho A_s V^2 D C_T \quad (3.2)$$

After the calculation of torque power produced by the rotor is needed to be calculated, formula used for the calculation of power is [32]:

$$P_{rotor} = T \times w \quad (3.3)$$

Power produced by the rotor is different than the power transferred to the blades via drag produced by wind, so power transferred by the air is [32]:

$$P_{max} = \frac{1}{2} \rho A_s V^3 \quad (3.4)$$

Ratio of both the powers gives the coefficient of power or efficiency which is [32]:

$$C_p = \frac{P_{rotor}}{P_{max}} \quad (3.5)$$

Aspect ratio plays an important role in the efficiency of turbine, and it is the ratio of height and diameter of blade and is [11]:

$$Aspect\ Ratio = \frac{H}{D} \quad (3.6)$$

Speed by which air strikes the blade is the ratio of rotor velocity and wind velocity which is [11]:

$$Tip\ Speed\ Ratio = \frac{U}{V} \quad (3.7)$$

There is also a relation between coefficient of torque and power which is [11]:

$$C_p = TSR \times C_T \quad (3.8)$$

3.4.2 H-Darrieus Turbine

The Power available in the wind is proportional to the cube of the wind speed and the area of the turbine A:

$$A = 2 R L \quad (3.9)$$

where R and L are the radius and length of the blade, respectively.

The total power available is given as

$$Power\ extracted = \frac{1}{2} \times \rho \times A \times v_w^3 \quad (3.10)$$

Where, ρ is the density of the air and v_w is the undisturbed wind speed. But like already known is the available power limited of the Bertz criteria. It is obvious that the power depends heavily on the wind speed.

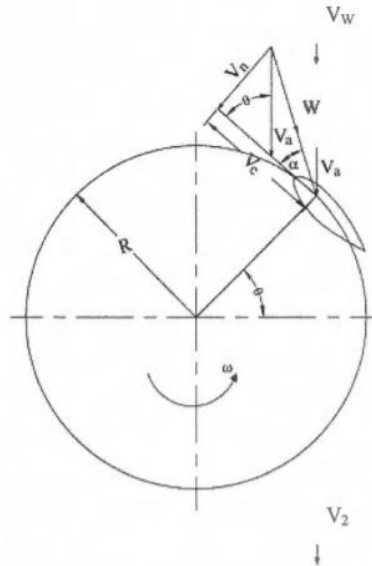


Fig 3-17: Free Body Diagram of Aerofoil[8]

The flow velocities in the upstream and downstream sides of the Darrieus type VAWTs are not constant. From this figure one can observe that the flow is considered to occur in the

axial direction. The chordal velocity component V_c and the normal velocity component V_n are, respectively, obtained from the following expressions:

$$V_c = R\omega + V_a \cos\theta \quad (3.11)$$

$$V_n = V_a \sin\theta \quad (3.12)$$

where V_a is the axial flow velocity (i.e., induced velocity) through the rotor, ω is the rotational velocity, R is the radius of the turbine, and θ is the azimuth angle. The angle of attack (alpha) can be expressed as,

$$\alpha = \tan^{-1} \left(\frac{V_n}{V_c} \right) \quad (3.13)$$

The directions of the lift and drag forces and their normal and tangential components. The tangential force coefficient (C_t) is basically the difference between the tangential components of lift and drag forces. Similarly, the normal force coefficient (C_n) is the difference between the normal components of lift and drag forces. The expressions of C_t and C_n can be written as

$$C_t = C_l \sin\alpha - C_d \cos\alpha \quad (3.14)$$

$$C_n = C_l \cos\alpha + C_d \sin\alpha \quad (3.15)$$

The net tangential and normal forces can be defined as

$$F_t = C_t \frac{1}{2} \rho \cdot f \cdot l \cdot V^2 \quad (3.16)$$

$$F_n = C_n \frac{1}{2} \rho \cdot f \cdot l \cdot V^2 \quad (3.17)$$

Where ρ is the air density, f is the blade chord, l is the length of the blades and V is the relative flow velocity, which is the velocity of wind.

Since the tangential and normal forces are for any azimuthal position, they are considered as a function of azimuth angle θ . Average tangential force (F_{ta}) on one blade can be expressed as, and the total torque [2] for the number of blades (N) is obtained as,

$$T = NF_{ta}R \quad (3.18)$$

And for the power [2],

$$P = T \cdot \omega \quad (3.19)$$

3.5 Details to Consider for Modelling of Helical Savonius Wind Turbine

A savonius turbine whose blades are twisted by some angle and is used to extract energy from wind is called Helical Savonius Wind Turbine. It usually has three blades mounted on a vertical shaft. Angle of twist of turbine ranges from 0 to 180 degree. There are number of parameters that effect the efficiency of helical savonius turbine:

- Angle of Twist
- End Plates
- Percentage Overlapping
- Number of Blades

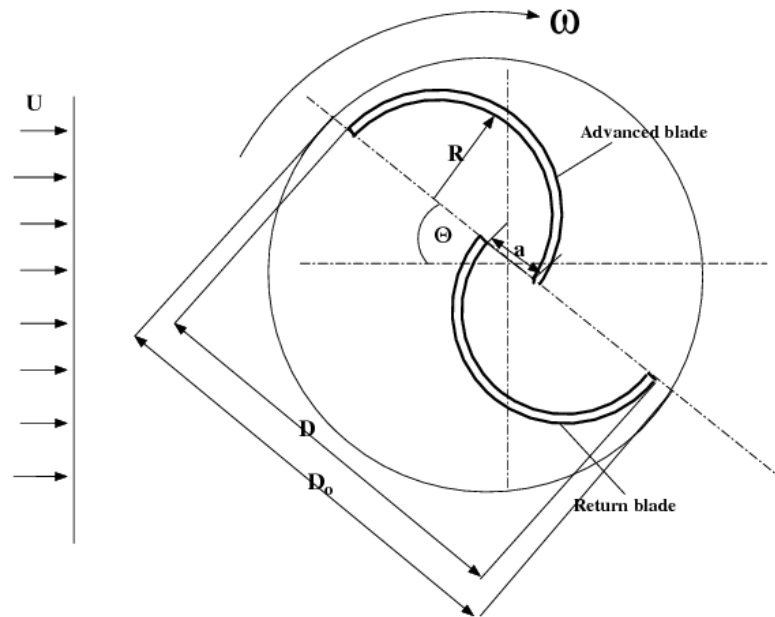


Fig 3-18: Figure showing parameters of VAWT [9]

Where, **a** represents overlapping, **D** is the diameter used in Calculating swept area, **D_o** is the diameter of endplates and **R** is the radius of blade

3.5.1 Angle of Twist

Angle of twist has a huge impact on efficiency of turbine, as angle of twist changes wind attacks the blade differently because of changed orientation of blade. This difference in attack is the main reason of change in impact. This impact varies at different angles. A simulative study was conducted by Lee et al., that study shows the power coefficient C_p at different angles. The angles they took into account were 0, 45, 90 and 135. Among these angles twist of 45 degree was giving the maximum efficiency.

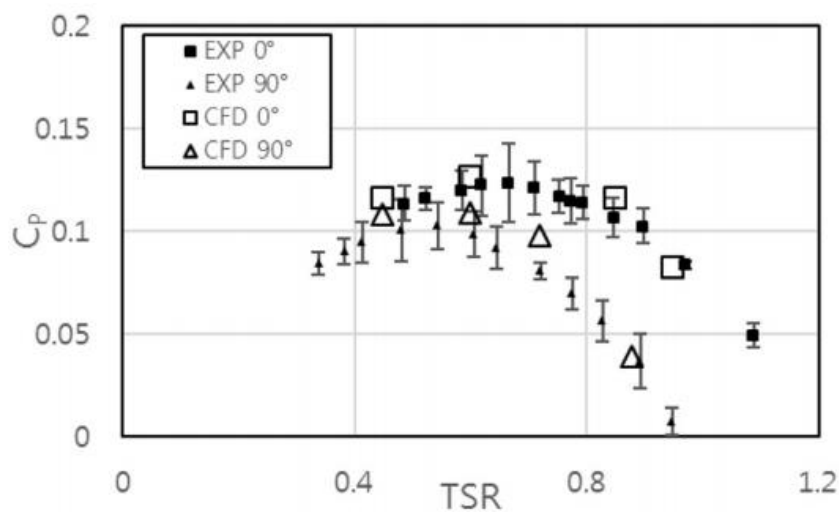


Fig 3-19: Impact of angle of twist 0 and 90 [10]

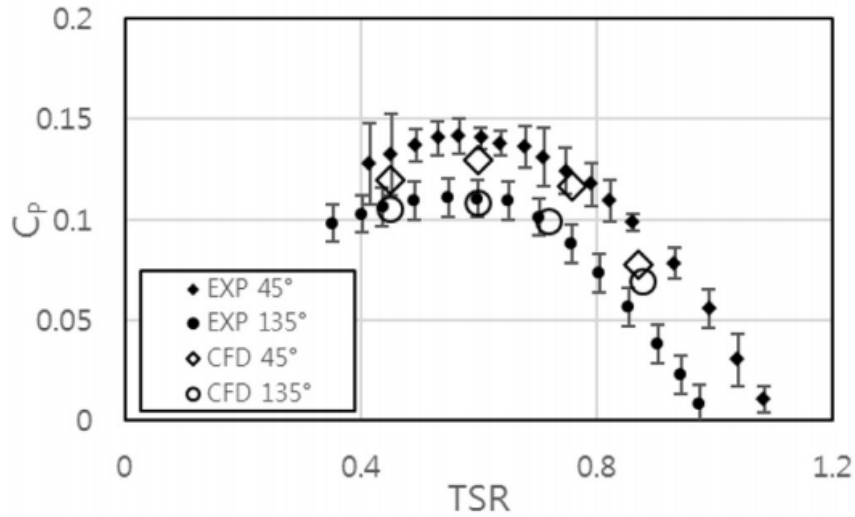


Fig 3-20: Impact of angle of twist 45 and 135 [10]

As shown in the graphs maximum efficiency lies between tip speed ratio of 0.4 to 0.8 for all the four angles but among those four angles maximum efficiency is at angle of twist of 45 degree. Blades with angle of twist of 45 degree also showed maximum efficiency when tested experimentally. Reynolds number used in simulative analysis was 1,440,000.

But this study didn't tell anything about twist of 180°. For that purpose, another study was conducted by Damak et al., that actually analyzes the power coefficient of turbine whose blades are twisted by 180 degrees at different Reynolds number.

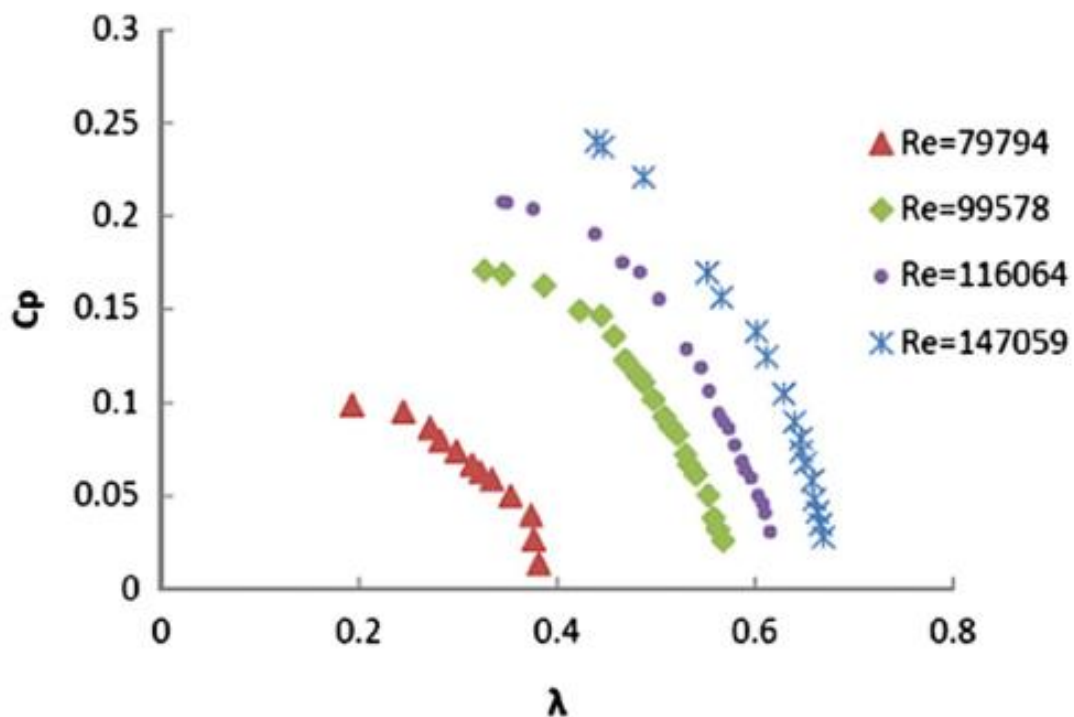


Fig 3-21: Impact of Reynolds Number [11]

As shown in graphs within specific range of tip speed ratio from 0.4 to 0.8, as it was in previous one, lies the maximum efficiency and maximum efficiency is at maximum Reynolds number of 147,059. At angle of twist of 180° more than 20% of efficiency can be achieved at Reynolds number 10 times lower than at angle of twist of 45° (as discussed in previous study).

3.5.2 Effect of Endplates

As the name suggests, plates that are applied at the end of blades are called endplates. These plates also have an impact on efficiency of turbine. Impact of these plates can be noticed experimentally or by simulation only, there is no theoretical backing like why it happens? To analyze the impact of endplates a study was conducted (3), in the present study comparison of 4 different types of endplates was done along with no endplate condition.

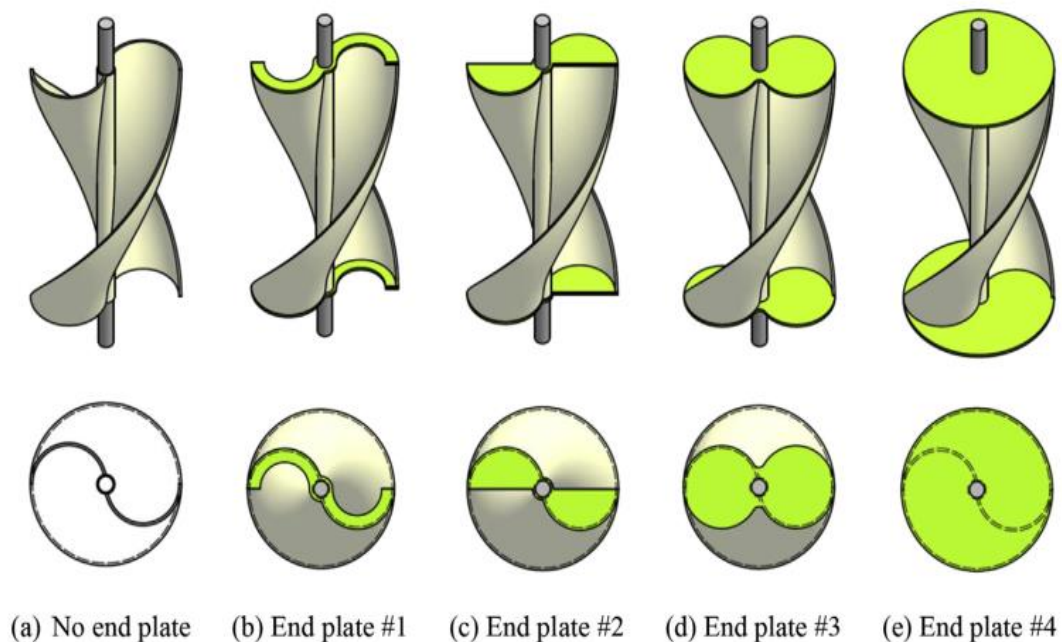


Fig 3-22: Types of End Plates [12]

Four different types of end plates are shown above. A simulative study was done of these plates and results were compared. Results showed that turbine with no end plate has less efficiency as compared to the turbine with end plates. But among the different types of end plates, the most effective is the one which covers whole blade. Analysis also showed that when end plates have covered the whole area as in end plate # 4, further increase in area of end plate leaves no impact on efficiency. Mathematically we can say when cross sectional area becomes equal to area of end plate, we get maximum efficiency or when their ratio becomes 1.

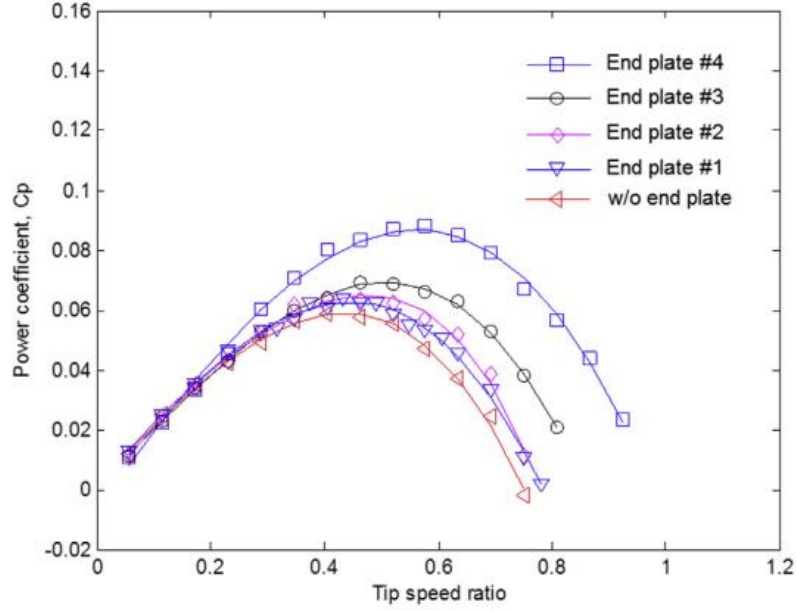
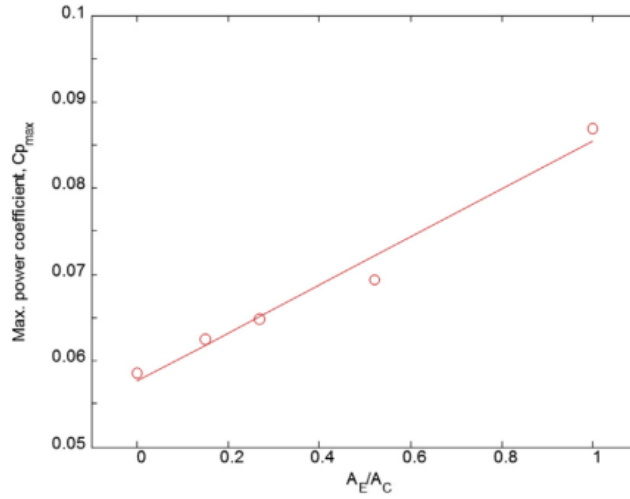


Fig 3-23: TSR vs Cp [12]



(a) The maximum power coefficient

Fig 3-24: Impact of End plates on Cp [12]

As shown in first graph maximum efficiency lies between 0.4 to 0.8 of tip speed ratio. In that range maximum power coefficient is of end plate # 4. 2nd graph shows as we increase the ratio of A_E (area of end plates) and A_C (cross sectional area), power coefficient increases, and maximum is at the ratio of one.

3.5.3 Percentage Overlapping

Overlapping is defined as the amount by which blades' diameters are overlapping each other, percentage of that overlapping is called percentage overlapping like how many percent of diameter are being overlapped. This overlapping also effects the efficiency of turbine. There is no direct relation that shows impact of overlapping, it can be found out

using simulation or by experimentation. For that purpose, a simulative study was conducted by Gupta et al., in that study turbine was passed through series of different overlapping percentages and results were compared.

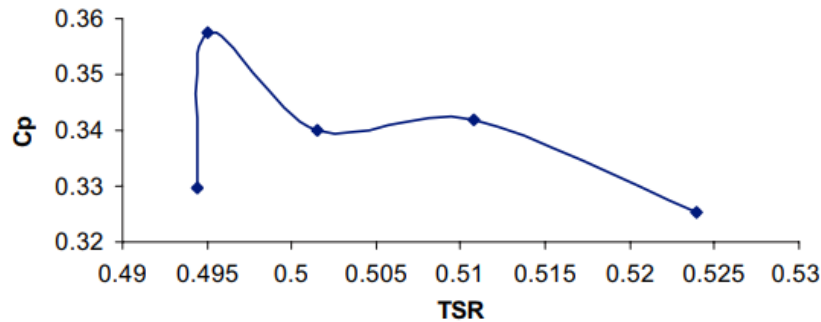


Fig 3-25: Cp without Overlap [13]

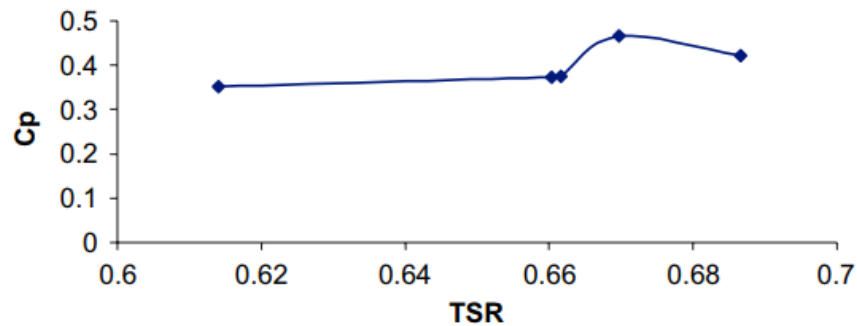


Fig 3-26: Cp with 20% Overlap [13]

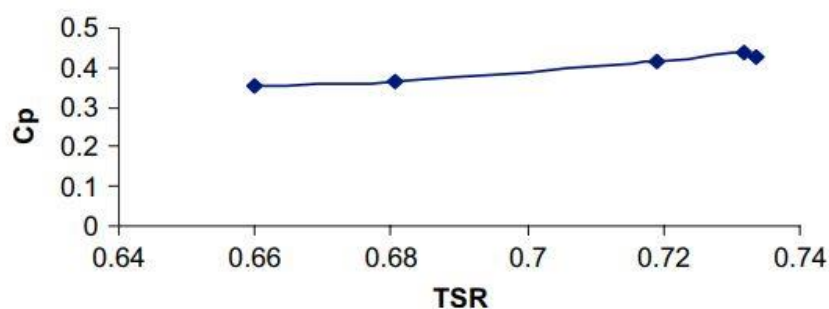


Fig 3-27: Cp with 16.2% Overlap [13]

Range of tip speed ratio is still same, as it can be seen that without overlap Cp goes up to maximum of 0.36 and it's not uniform too there is too much fluctuation. For 20% overlap maximum Cp is up to 0.5 but that value is for specific tip speed ratio and Cp also experience a sudden increase. While for 16.2% overlap maximum Cp is between 0.4 and 0.5 and increase in Cp is uniform too.

3.5.4 Number of Blades

Number of blades has a huge impact on efficiency of turbine, to investigate that a study was conducted by Rashidi et al, in that study helical savonius wind turbine with 2, 3 and 4 blades was tested under same conditions, and it was concluded that three number of blades are the best option.

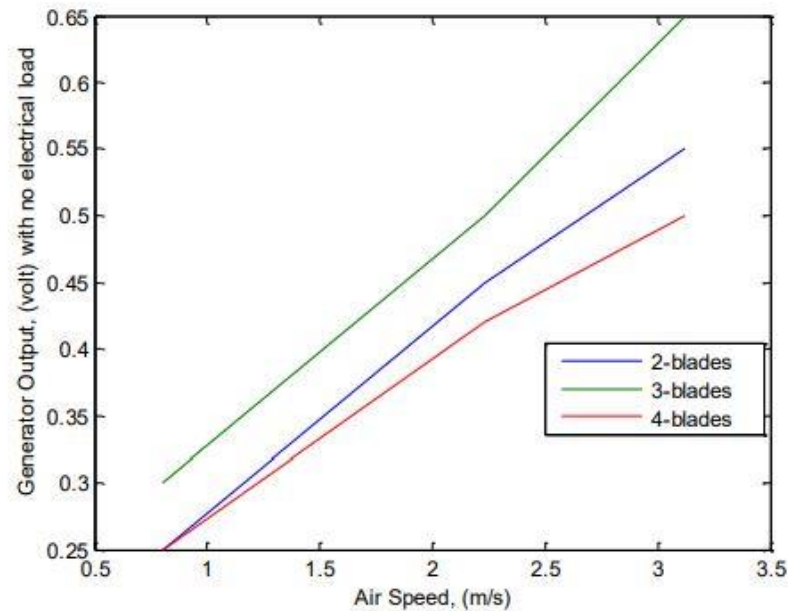


Fig 3-28: Impact on no. of blades [14]

3.5.5 Calculations

Calculations on the basis of formulas given in above section.

Assuming the diameter and velocity of blade to be:

$$D = 0.3m$$

$$V = 12 \text{ m/s}$$

$$AR = 2$$

So,

$$H = 0.6m$$

Swept area of blades will be:

$$A_s = \pi DH = 0.519 \text{ m}^2$$

Maximum power or power transferred by wind is:

$$P_{max} = 151.34 \text{ W}$$

3.5.6 Geometric Modeling

Geometric modeling was done in Creo 6.0. Following steps were followed for modeling in Creo:

- Selection of Turbine
- Considering all the Parameters Above
- Calculations for Height and Blade
- Sketching in Creo
- Generating 3D Model

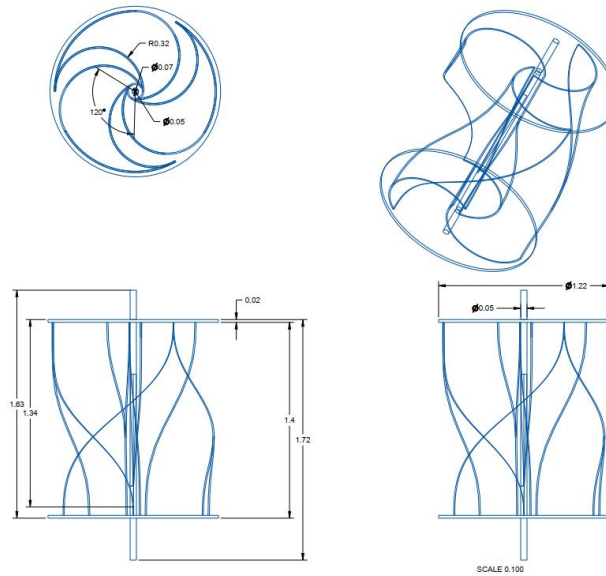


Fig 2-29: 3D Model of Helical Turbine

3.6 Details to Consider for Modelling of H-Darrieus

- Type of Airfoil
- Struts Distance or H/D Ratio

3.6.1 Airfoil Selection

Cross-sectional shape of an object which generate aerodynamic force while moving through fluid is known as aero foil. Due to their small size and ability to perform well even in the face of low-speed and unstructured flows with low noise emissions and good durability, Whether VAWT is drag or lift based, are gaining popularity these days, especially in medium and small-scale installations. H-Darrieus rotors, for example, are becoming more popular in unusual applications because they are thought to improve performance even when the oncoming flow is misaligned with the rotor's axis. Industrial manufacturers are developing new and more appealing design solutions on the one hand, and efforts are being made to reduce the initial cost of the machines (primarily through new materials) and increase their efficiency in order to make them competitive with more traditional Horizontal Axis Wind Turbines on the other hand, in order to promote the adoption of this technology (HAWTs).

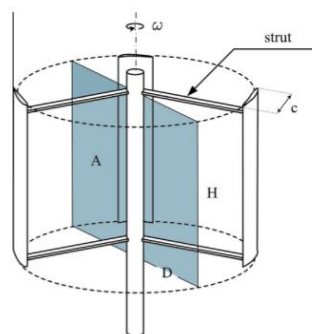


Fig 3-30: H-Darrieus[8]

Aerofoil Selection is an essential phase for the design of an H-Darrius wind turbine as the VAWT is essentially based on the drag concept. Aerofoil with greater drag within the given condition is selected for the final design as much power is produced due to the greater drag on its blades compared to the other aerofoils considerations. An aerofoil determines the lift and drag coefficients, in this context three aerofoils were analysed, which are mentioned as.

- NACA0015
- ClarkY
- FX-74-C15-140 (Smoothed)

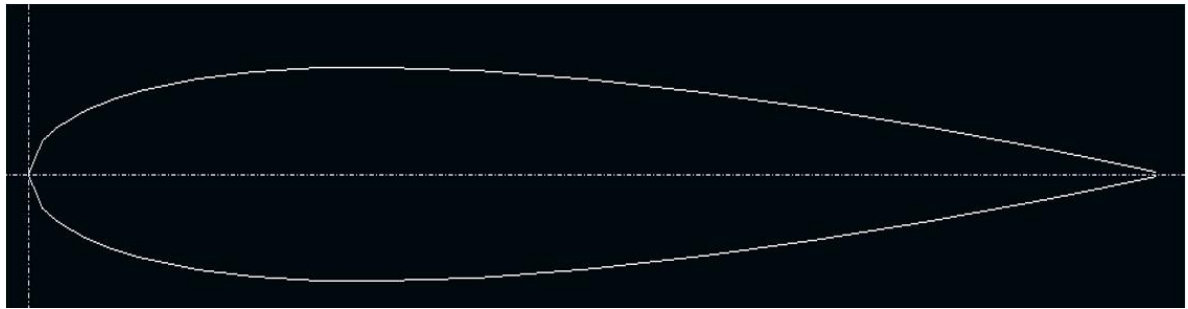


Fig 3-31: NACA0015

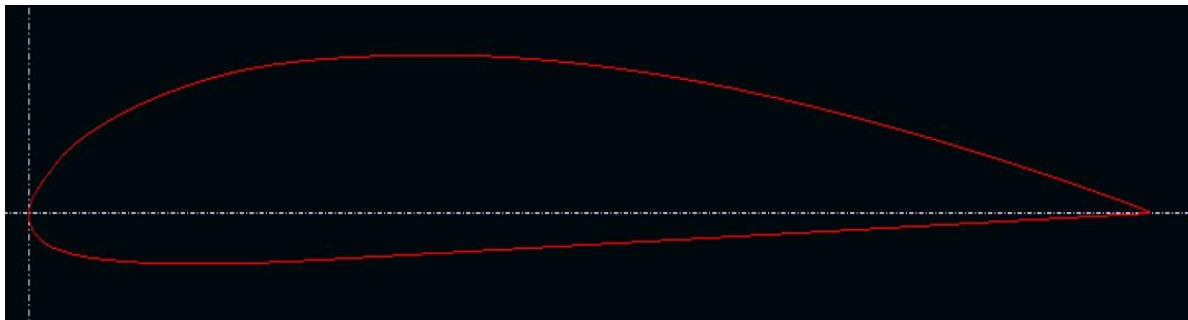


Fig 3-32: ClarkY

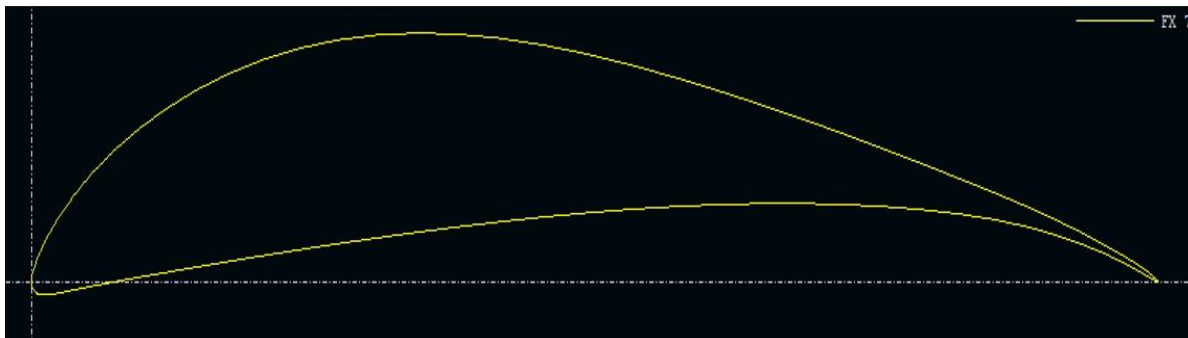


Fig 3-33: Smoothed FX-74-C15-140

NACA0015 is a symmetric aerofoil while FX-74-C15-140 (Smoothed) is high lift aerofoil due to its lower frontal area. ClarkY is an approximately a flat bottom aerofoil which is in

between the shape of NACA0015 and FX-74-C15-140. The Aerofoil selection was based on the following parameters.

- C_l to α ratio
- C_d to α ratio
- C_l to C_d ratio
- Ease of fabrication

The analysis was done using xflr5 (online software) for the Re of 200,000 and α ranges from 1 to 20 degrees, the results are as in the case of VAWT, angle of attack remains within 20 degrees.

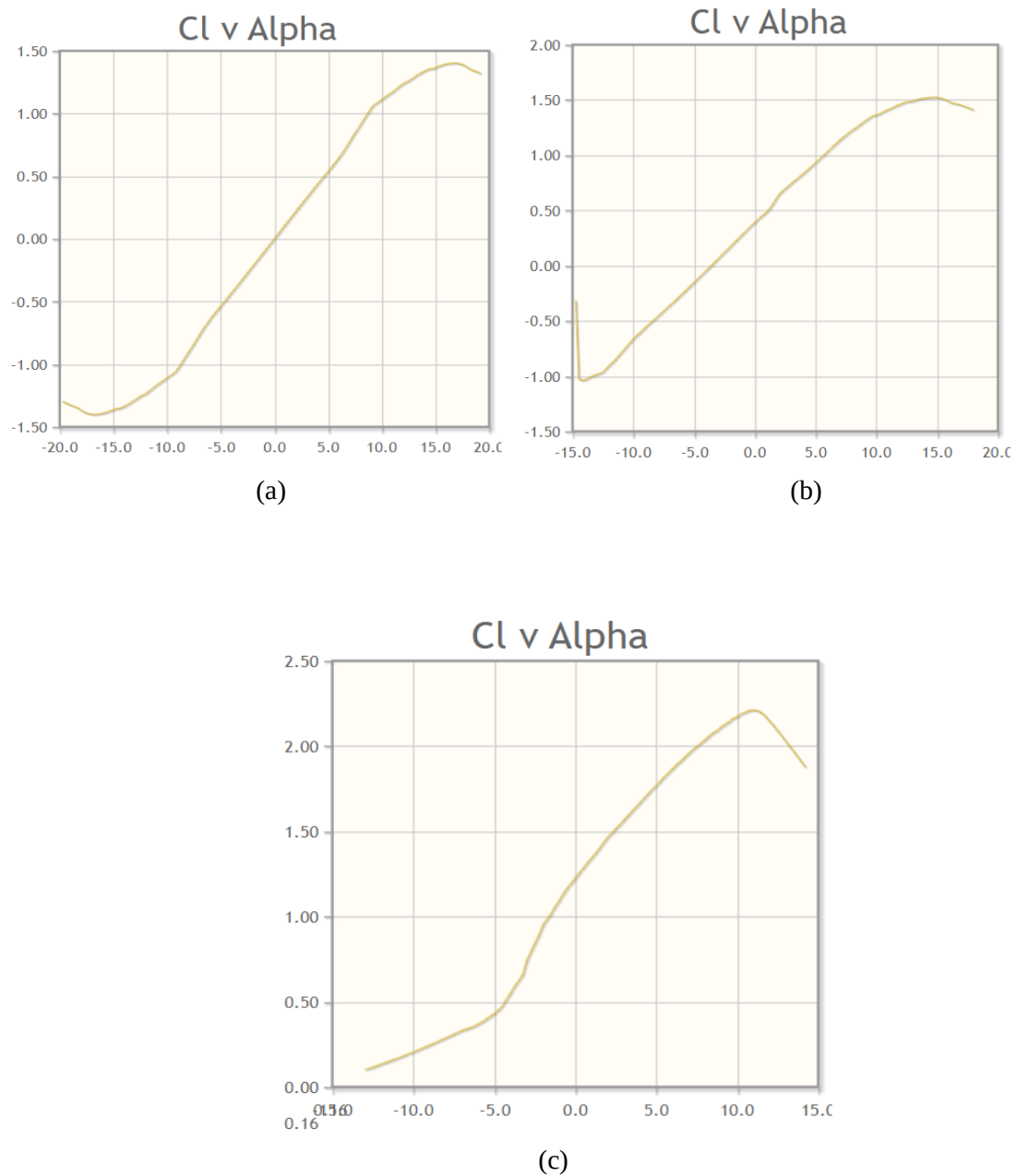


Fig 3-34: C_l vs α a) NACA 0015 b) ClarkY c) FX-74-C15-140

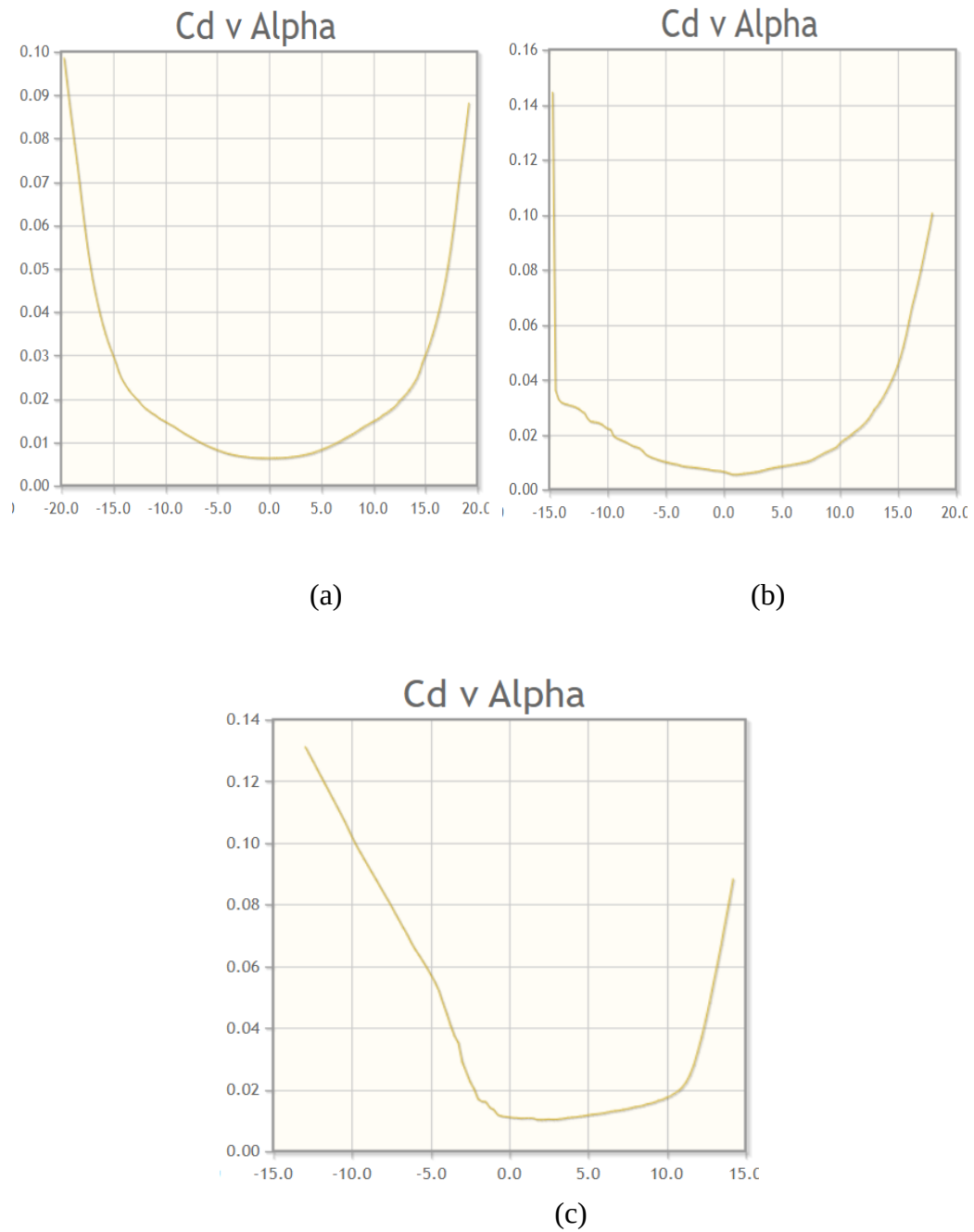


Fig 3-35: Cd vs Alpha a) NACA 0015 b) ClarkY c) FX-74-C15-140

As shown in the figures, it has been seen that angle of attack has the great impact on the lift and drag co-efficient. For the optimization of lift to drag ratio, the final model has the one with high. For the final model, analysis from these graphs has been shown that NACA0015 has the optimized lift to drag ratio at greater than 15 degrees. Other two has also the optimized one but the ease of fabrication compels to go with the NACA0015, as the achieved results from the other two types of airfoils lies within the range of it.

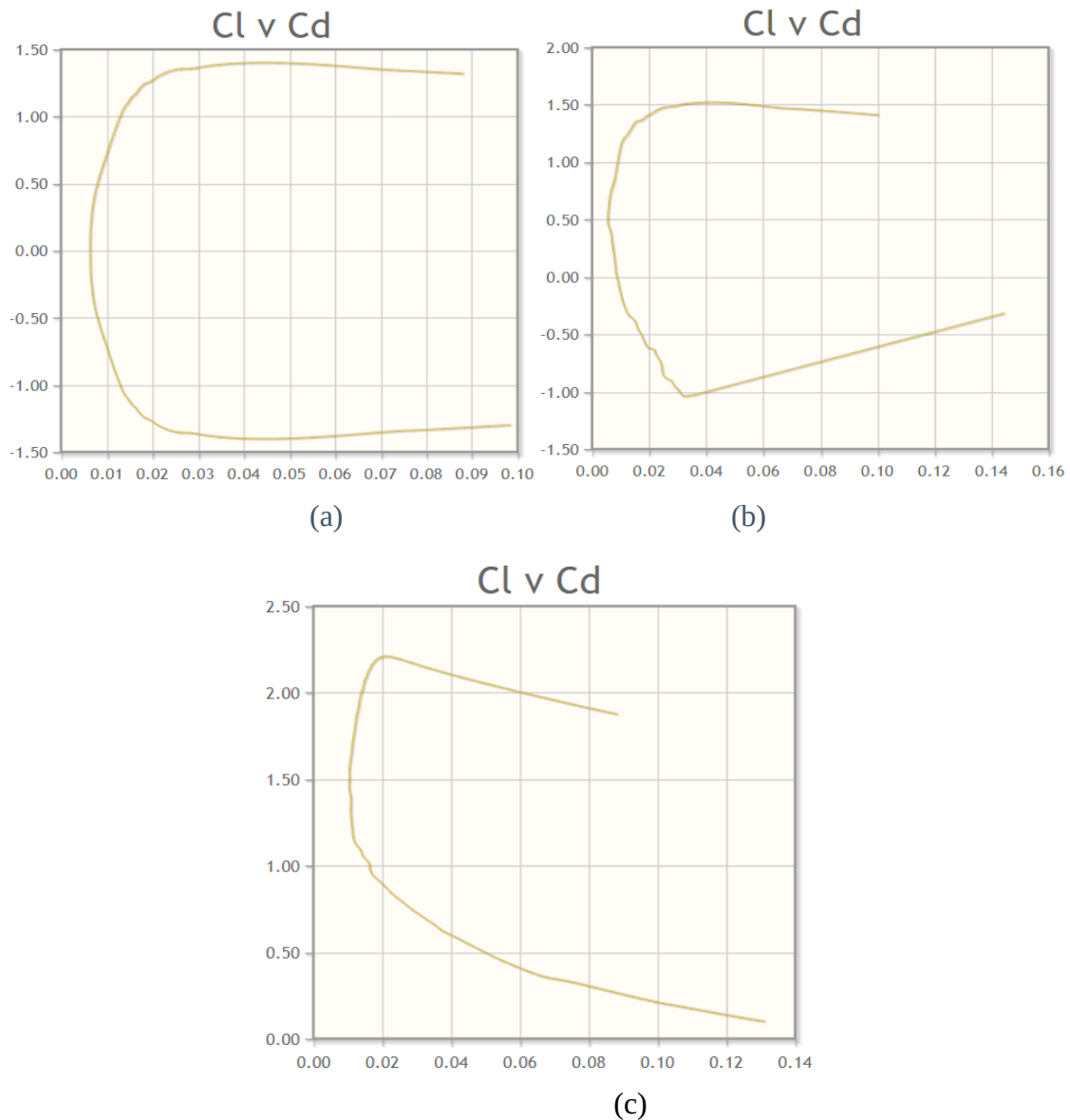


Fig 3-36: C_l vs C_d a) NACA 0015 b) ClarkY c) FX-74-C15-140

3.6.1.1 Analysis from the Graphs

3.6.1.2

Following information was obtained from the above graph.

- NACA0015 has the higher value of drag coefficient for the given range of angle of attack followed by ClarkY and FX-74-C15-140 (Smoothed), respectively.
- FX-74-C15-140 (Smoothed) has the higher value of lift coefficient for the given range of angle of attack followed by ClarkY and NACA0015, respectively.
- The Lift and Drag Coefficient were compared for the given range of angle of attack.
- NACA0015 has also easy to fabricate than ClarkY and FX-74-C15-140 (Smoothed).

3.6.1.3 Final Selection

NACA0015 aerofoil was selected by the group for the following reasons.

- Highest value of drag coefficient for the range of angle of attack.
- Though its drag coefficient were less for lower degrees of angle of attack than that of FX-74-C15-140 (Smoothed) and ClarkY but due to ease of fabrication and overall design it was advised by the group to go NACA0015.

3.6.2 H/D Ratio

After the selection of an aerofoil the main concerns that arises is that how much is required distance for an aerofoil from the axis of rotation in order to keep the moment of inertia as minimum as possible, so that the VAWT is able to produce power at low wind speed.

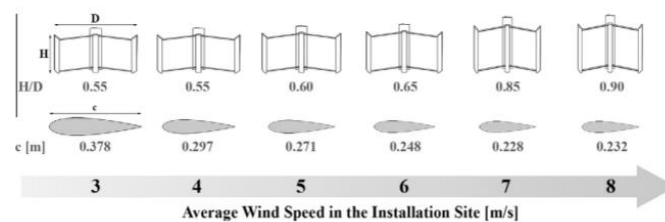


Fig 3-37: H-Darrius H/D Ratio[8]

It shows that if the H/D is increased, it should be installed at the sites where wind speed is high. For the H-Darrius VAWT the minimum H/D ration is 0.55 so that can be able to operate on the speed of 3m/s which is the suitable range for the site area.[1]

3.6.3 Calculation

From all the discussion above calculations were performed in the form of iterations.

3.6.3.1 Iteration # 1

Assumptions,

$$\text{Velocity of the wind} = 10 \frac{\text{m}}{\text{s}}$$

$$\rho = 1.125 \text{ kg/m}^3$$

$$\text{Chord length} = f = 0.12 \text{ m}$$

$$\text{Blade length} = l = 0.5 \text{ m}$$

We have given the wind speed, but we must have to analyze our turbulence model. After the selection of an aerofoil, turbulence model with the angle of attack, Reynolds number and the angle of attack is important,

$$\text{Re} = 50,000$$

Assume the angle of attack to be 10^0 .

$$\alpha = 10^0$$

The Ncrit value is a measure of free flow turbulence and is used to simulate the transition location when no forced trip location is given. For the different Ncrit value, we are obtaining a different graph which are,

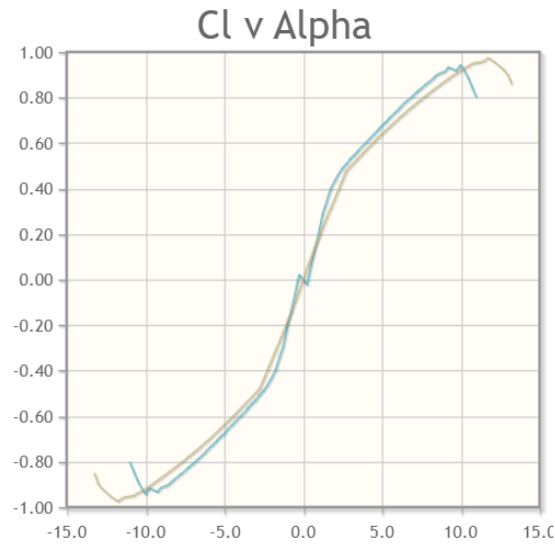


Fig 3-38: Cl vs Alpha graph [15]

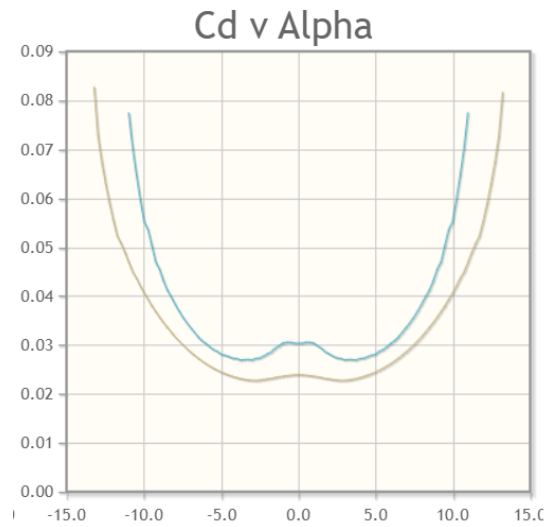


Fig 3-39: Cd vs Alpha graph[15]

$$C_l = 0.90$$

$$C_d = 0.06$$

$$C_t = C_l \sin \alpha - C_d \cos \alpha = 0.145$$

$$C_n = C_l \cos \alpha + C_d \sin \alpha = 0.1667$$

Now for the tangential and normal forces on the aerofoil is.

$$F_t = C_t \frac{1}{2} \rho \cdot f \cdot l \cdot V^2 = 0.489 \text{ N}$$

$$F_n = C_n \frac{1}{2} \rho \cdot f \cdot l \cdot V^2 = 0.5626$$

$$\frac{H}{D} = 0.5$$

$$D = 0.35$$

$$T = NF_{ta}R = 0.0776 \text{ N.m}$$

3.6.4 Geometric Modeling

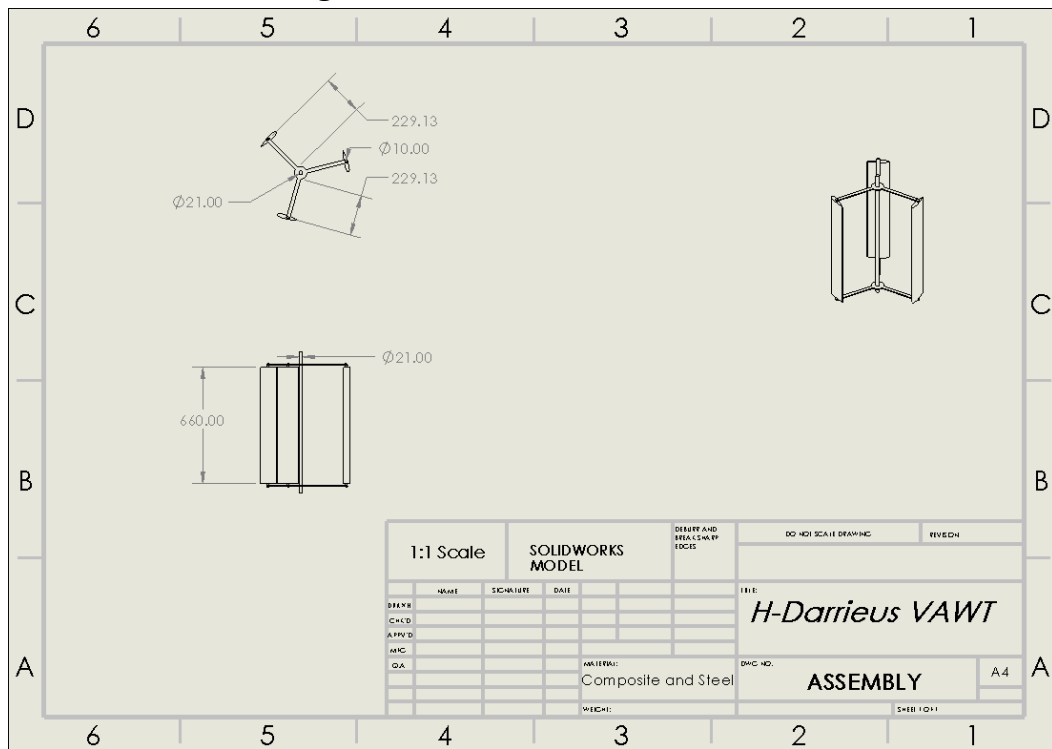


Fig 3-40: Drawing of H-Darrieus

3.7 Shaft Design

To minimize weight, hallow shaft has been considered.

3.7.1 Assumptions

Table 3-1: Assumptions for shaft design

Final Model	
Material	Structural Steel
Max Velocity	7.5 m/s
Safety Factor	3
Density of air	1.125kg/m ³
Outer diameter	25mm

3.7.2 Calculated terms

Swept Area =0.96m.

$$d = 0.6 \text{ m}$$

$$T = \frac{1}{4} \rho_w A_s d v_w^2 \quad (3.20)$$

$$T = 9.09 Nm$$

As

$$T_{max} = S.F \times T \quad (3.21)$$

Torsion Analysis

$$\tau_{max} = \frac{Tr_2}{\frac{\pi}{2}(r_2^4 - r_1^4)} \quad (3.22)$$

Outer diameter= r_2 =25mm

Thus, shaft with outer diameter of 25 would be a great a choice for physical model.

3.8 Spur Gears

Spur gears have teeth parallel to the axis of rotation and are used to transmit motion between parallel shafts.

3.8.1 Designing a Gear

3.8.1.1 Assumptions

To start off we assume a few properties. These include:

Module[17] = 2mm

Pressure angle(ϕ) = 20°

No. of teeth on pinion [31] = 16

No. of teeth on pinion [4] = 32

3.8.1.2 Calculations [33]:

Center distance:

$$C = \frac{m(N_P + N_G)}{2} \quad (3.23)$$

Pitch Diameters:

$$d_G = mN_G \quad (3.24)$$

Circular Pitch[2]:

$$p = \pi m = 2\pi \text{ (Same for all gears)} \quad (3.25)$$

Base Diameters:

$$d_{bp} = d_p \cos \phi \quad (3.26)$$

Addendum[1] and Dedendum[1]:

$$a = 1 \times m \quad (3.27)$$

$$b = 1.25 \times m$$

Tooth Thickness:

$$t = \frac{p}{2} = \frac{3\pi}{2}$$

Clearance:

$$c = b - a$$

Base Pitch:

$$p_b = p \cos \phi$$

Interference:

We will assume full depth teeth so $k=1$

$$N_p = \frac{2(k)}{(1+2m)(\sin(\phi))^2} (2 + \sqrt{m^2 + (1+2m)(\sin(\phi))^2})^2 \quad (3.28)$$
$$= 14.16 \approx 15$$

This is the smallest teeth on pinion that will be interference free. Hence, the pinion is without interference.

$$N_G = \frac{N_p^2 \times (\sin(\phi))^2 - 4(k)^2}{4(k) - 2(P_p)(\sin(\phi))^2} = 102 \quad (3.29)$$

Parameters	Values	Parameters	Values
Center distance	48 mm	Base pitch	8.86mm
Pitch diameter (Pinion)	32	Pitch diameter (Gear)	64
Circular diameter	2π	Clearance	0.75
Addendum	2 mm	Dedendum	2.5 mm
Tooth thickness	4.71 mm	Base diameter (Gear)	4.51 mm
Base diameter (Pinion)	4.51 mm		

This is the largest number of teeth on gear which will be interference free with the specified pinion. Hence our gear is within the interference free limit.

3.9 Bearing Selection

As angular contact bearing provides greater speed so angular contact bearing is selected for the calculation.

From table 11.2 (**Table 3-2**) Bearing specifications [33]

Parameters	Values
Bore	40mm
Outer Diameter	80mm
Width	18mm
C_0	18.6KN
C_{10}	31.9KN

Assuming the desired life to be equal to rating life:

$$F_i = \frac{0.47F_r}{k} \quad (3.30)$$

F_r is the reaction force and k is a constant whose value is $k=1.5$.

$$F_{eA} = 0.4 \times F_{rA} + K_A \times (F_{iB} + F_{ae}) \quad (3.31)$$

$$F_{eB} = F_{rB}$$

Where F_{eA} and F_{eB} = the dynamics equivalent loads,

F_{ae} =Thrust load (mainly due to the weight of the blade and shaft

F_{rA} and F_{rB} = radial loads at A and B Respectively.

Multiple rating of life

$$X_D = l_D \times n_D \times \frac{60}{l_R} \quad (3.32)$$

$$C_{10} = (a_f) \times F_D \times \left(\frac{x_D}{(x_0 + (\theta - x_0) \times (1 - R_0))^{\frac{1}{b}}} \right)^{\frac{1}{a}} \quad (3.33)$$

For angular contact bearing

Application Factor	3	L_{10}	10^6 revolutions
R_d	0.99	X_0	0.02
$\theta-x_0$	4.43	b	1483

$$\frac{C_{10}}{\left(\frac{x_D}{(x_0 + (\theta - x_0) \times (1 - R_0))^{\frac{1}{b}}} \right)^{\frac{1}{a}} \times (a_f)} = F_D \quad (3.34)$$

Final Design	
F_D	2656N

Now

As weight = 550N

$F_a=550N$

$$F_r=550 \times 0.2 = 110N$$

Now using Equation 11-9

$$F=XVF_r+YF_a$$

Here $V=1$ because inner ring rotates.

Thus from table 11-1

$$X_1=0.56 \text{ and } Y_2=2.30$$

$$F=1326.6N$$

$$S.F = \frac{2656.6}{1326.6} > 2.003$$

As factor of safety is greater than 2, thus this bearing can be used in the final design.

3.10 Design codes and standards

Table 3-3: Design code and standards

Design codes/Standards	Design Factor	Criteria
ISO/IEC 9126-1,	Strength	Withstand factored loads and fatigue loads
ISO/IEC 9126-1	Stability	Remain under the extreme loads
ISO 743	Flexural Failure (blades and shafts)	Resist failure due to high bending moments.
ISO 6336, ISO 5833	Axial and bending loads (Bearings)	Considers the combined failure due to axial and bending moment.

3.11 Chapter summary

This chapter details the design process of the multistage wind turbine. First of all the design methodology has been discussed after which process flow chart has been shown. Mathematical modeling of both rotors has been done using several iterations and the best design iteration is chosen for the final model. The selection of an aerofoil for H-Darrieus and angle of twist for helical savonius were the key steps. After numerical modeling for rotor, analytical technique has been explained for the selection of shafts, bearings and gears. At the end of the chapter design codes and standards are discussed.

Chapter 4

DEVELOPMENT PROCESSES

4.1 Fabrication of a prototype

After selecting the design for the highway turbine, it has been decided to fabricate a prototype to check and verify the simulation results. At first it was hoped to 3D print the complete prototype. But it was inaccessible to print the complete model as the cost would have also been increased marginally and also the results obtained will not be comparable to the simulation due to the difference in material properties. Thus, it had been decided the fabricate the model by using the material that had been used during the simulation.

4.2 Fabrication Helical Savonius

Helical savonius provides the better results in terms of starting torque. First of all, it was decided to achieve the desired angle of the blades by melting down the PVC pipe but failed to achieve. After failing in first attempt, second attempt was made. In this attempt, PVC pipe was replaced with shuttering sheets. It was tried to achieve the desired helical blade angle manually, but it fell short. After failing of second attempt, final attempt was made which was 3D printing of the blade. During the printing of a blade, fault occurred in the nozzle of the available 3D printer. When the third attempt was crashed, it had been decided to review the proposed design of the turbine as other techniques were quite complex. By reviewing, it had been decided to replace helical savonius turbine with simple savonius turbine due to ease of fabrication and the results provided by the savonius turbine are close to helical savonius turbine.

4.3 Updated proposed model

As the it has been decided to replace the helical savonius turbine with the savonius turbine. Savonius turbine provide ease in the fabrication due to its simple structure. Simple straight semi-cylindrical blades are the key factors which differs the structure of savonius turbine from helical savonius turbine. The following Fig 4-1. shows the 3D model of the modified proposed design.

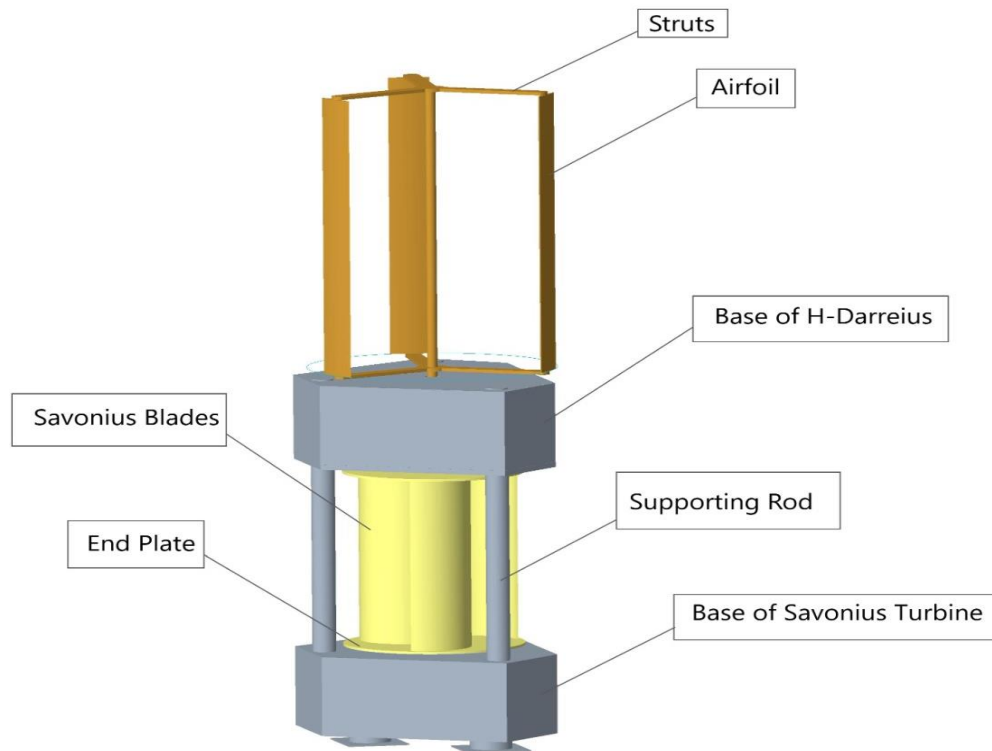


Fig 4-1: Updated proposed design

4.4 Fabrication of H-darrieus prototype

Fabrication of H-darrieus turbine was started by making the airfoils of the turbine. Initially, NACA0015 were selected among the three types of air foil, and it was made of wood, but the results was no sufficient for our desired requirements as the thickness was not sufficient and the lift produced by that air foil was not enough. Thus, to improve the results hollow airfoils were made using mild steel thin sheet. The struts were made of mild steel. Once the blades were fabricated, other components like shafts, bearings, dynamo had been purchased according to the calculations. Base and retainers were also made by using lathe and milling machines. Finally, the assembly of all parts have been performed. Height to diameter ratio for the H-darrieus prototype is 0.55.



Fig 4-2: H-Darrieus Prototype

The following table details the components list used in the fabrication of H-Darrieus Turbine.

Table 4-1: Components list used in the fabrication of H-Darrieus Turbine

Components/Parts	Quantity	Properties	
		Type	NACA0015
Aerofoil	3	Material	Mild Steel
		Length	15 cm
Struts	6	Thickness	2 mm
		Material	Mild steel
Retainers	2	Outer diameter	2.5cm
		Type	Angular contact ball bearing (7004)
Bearings	1	Inner diameter	2cm
		Outer diameter	2.6cm
Dynamo	1	Input speed	Up to 5000rpm
		Max load Current	680mA
Shaft	1	Voltage	12V,6V
		Length	25 cm
Plates (Base)	2	Diameter	5 mm
		Material	Mild steel
Base rods	3	Shape	Triangular
		Dimensions	25cm all sides
		Material	Mild steel
		Diameter	2 cm
		Length	7 cm

4.5 Fabrication of Savonius turbine

Savonius turbine's fabrication was started by purchasing plastic pipe of for the blades. To attach the shaft, End plates were connected to the blades using glue. Once the rotor has been fabricated, bearings and dynamo were purchased. Iron plates were cut to form a base and top. Three supporting rods were used to hold the top. Rotor and base are assembled using retainers at both end of the shaft. Overlapping ratio for the blades were kept being 20% and height to diameter ratio is 3.

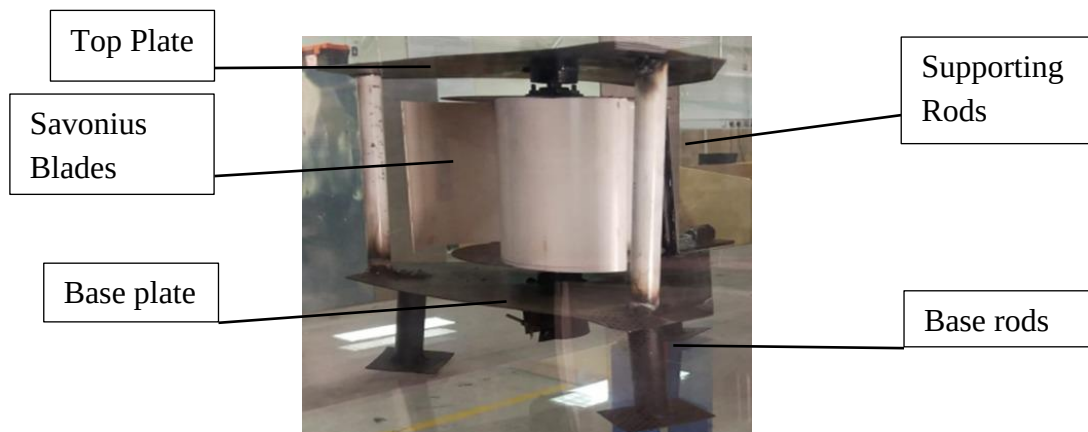


Fig 4-3: Savonius prototype

Table 4-2: List of the components used in the fabrication of Savonius turbine.

Components/Parts	Quantity	Properties	
Blades	3	Material	PVC
		Diameter	8cm
		Length	20cm
Shaft	1	Length	25cm
		Diameter	2mm
Endplates	3	Material	Mild steel
		Diameter	16cm
Supporting rods	3	Material	Mild steel
		Length	22cm
Dynamo	1	Input speed	Up to 5000rpm
		Max load Current	680mA
		Voltage	12V,6V
Bearings	2	Type	Angular contact bearing 7004
		Inner diameter	2cm
		Outer diameter	2.6cm
plate (Base and top)	2	Material	Mild steel
		Shape	Triangular
		Dimensions	25cm all sides
Base rods	3	Material	Mild steel
		Diameter	4mm
		Length	8 cm
Retainers	1	Material	Mild steel
		Outer diameter	2.5cm

4.6 Testing in Wind Tunnel

Once the fabrication phase of the prototype had been completed, testing in the wind tunnel has been carried out to check the feasibility of the turbine and validate the simulation results. The prototype was placed inside the wind tunnel and readings were taken by varying wind velocity up to 8 m/s. Angular velocity of the turbine was measured by the help of tachometer while voltage and current were measured using digital multi-meter (DMM). Torque coefficient was calculated analytically followed by torque and power. The results are discussed in the next chapter.

4.7 Fabrication of full-scale model

After successful attempt of prototype, the next was to fabricate the full-scale model.

4.7.1 Fabrication of H-Darrieus turbine

H-Darrieus turbine was selected to be used above the savonius turbine in the multi-staging process. It consists of three air foils, struts, shaft, base, and an DC generator.

The development process began with the fabrication of the air foils. By considering the results and experience gained by the prototype, it was decided to fabricate the NACA 0015 air foil which is symmetrical in shape, having thickness 8cm and chord line 34 cm, as it provides the better results for rotational motion due to optimum lift provide by the air foil. The available jumbolon board sheets were of thickness 5cm which is maximum. Thus, two sheets were stick together using epoxy resin and hardener solution. Once both sheets were stick together, two wooden moulds were made, which were glued to top and bottom of the sheet. These wooden moulds provided a boundary line, of the desired NACA 0015 air foil, for cutting process. Hot wire cutting was performed to cut out the NACA0015 air foil from jumbolon sheet. Nichrome wire was used for the hot wire cutting process as it has the perfect balance of energy requirements and tensile strength. DC current for the hot wire cutting was provided through a manually build electrical circuit. Once the cutting was done, filing was performed to furnish the surface of the air foil and smoothens the outer surface so that the maximum lift will be obtained and delayed the boundary layer separation. After furnishing the surface, three rectangular slots were made in the air foil where small wood rectangular blocks were placed as shown in figure 4-4. The wooden blocks were stuck to with the air foil using epoxy resin and hardener solution. The purpose of these blocks was to bolt down the struts with the air foil.

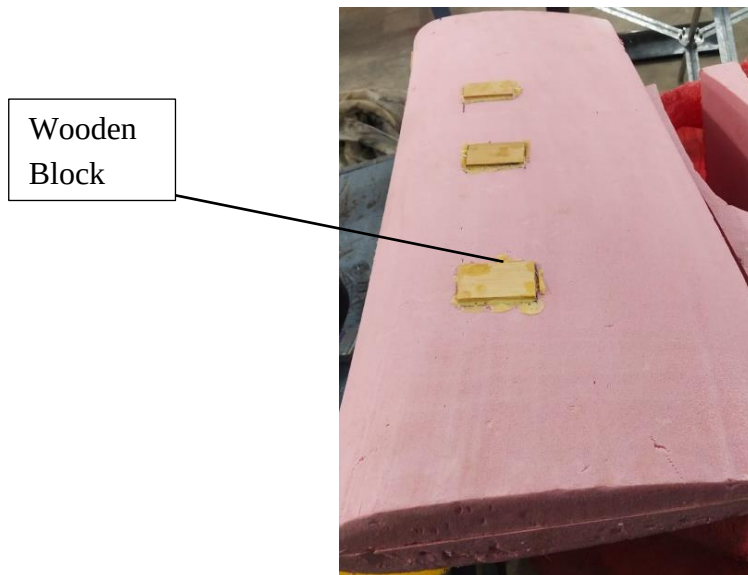


Fig 4-4: Air foil

Once the fabrication of air foil has been finalized, the next step was to make struts. Struts were made using mild steel rectangular bars as shown in Fig. 4-5. Each air foil was connected to two struts (upper and lower) shown Fig. These struts were connected to the respective retainers using temporary joint (Nuts and Bolts) as shown in Fig 4-6. Main shaft was inserted in the retainers. The top side of the shaft kept free while bottom side of the shaft was inserted into the retainer. This retainer is mounted on the base and has two bearings to keep the shaft in place as shown in Fig 4-7. One bearing is above the base while the other one is below the base plate. The purpose of this big retainer is to prevent the rotor of H-Darrieus from wobbling. Moreover, for added the strength of structure and maintain the alignment, upper end of the shaft got more free length. Base plate, having bended edges, is made up of mild steel sheet. Dynamo is placed under the base and connected to the shaft to transform rotational kinetic energy of the shaft into electrical energy.

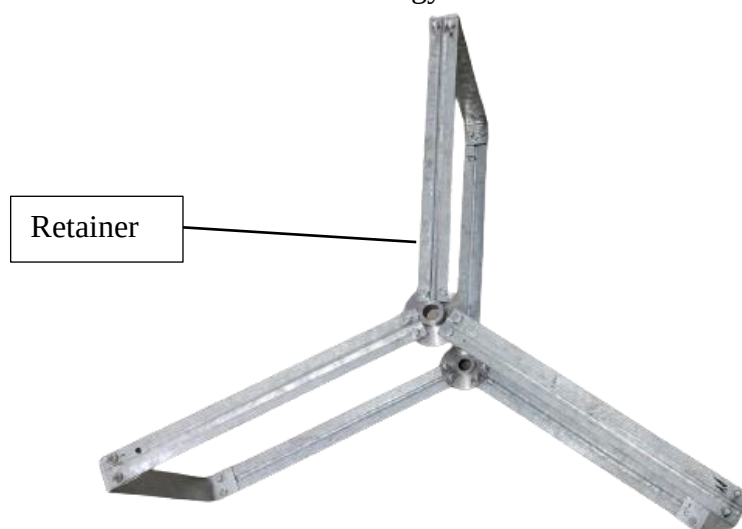


Fig 4-5: Struts Assembly



Fig 4-6: Strut attached with air foil

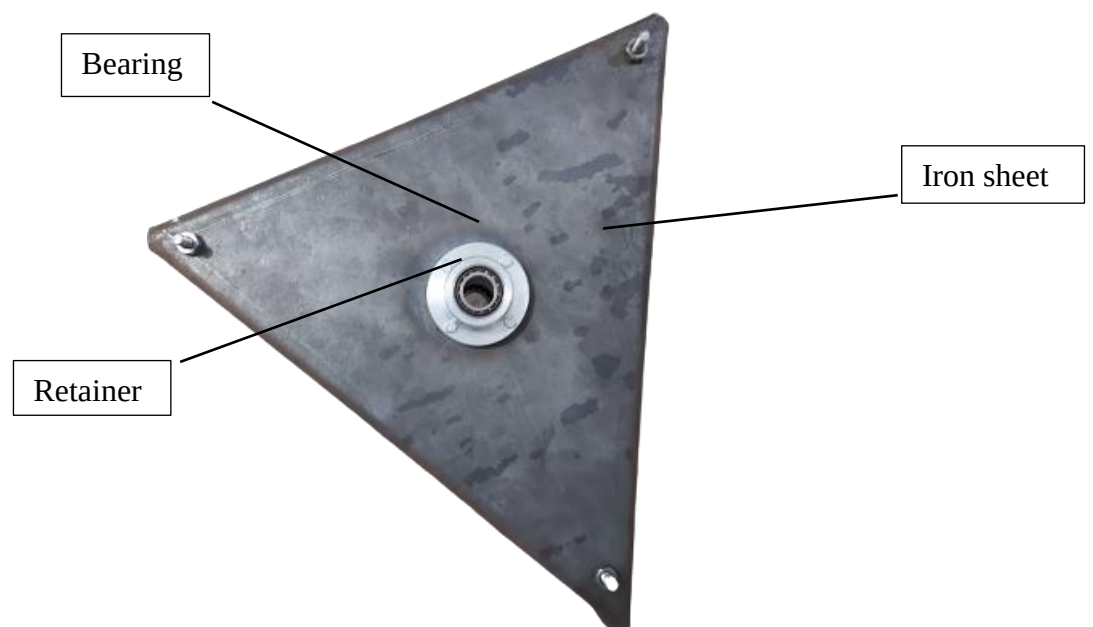


Fig 4-7: Base of the H-darrieus turbine

Assembly process of the H-Darrieus turbine began by connecting the struts to the retainers. After which shaft was inserted in the retainers. After connecting shaft to the struts, air foils are bolted to the struts via small wooden rectangular blocks. Once the rotor assembly has been completed, to connect the rotor with stator, shaft of the rotor has been pushed into the main retainer which is mounted on the base of the H-Darrieus turbine. Finally, the dynamo has been connected to the shaft to generate electricity. The H/D ratio remained same as of prototype that is 0.55. The following figure 4-8 shows the H-darrieus model assembly.

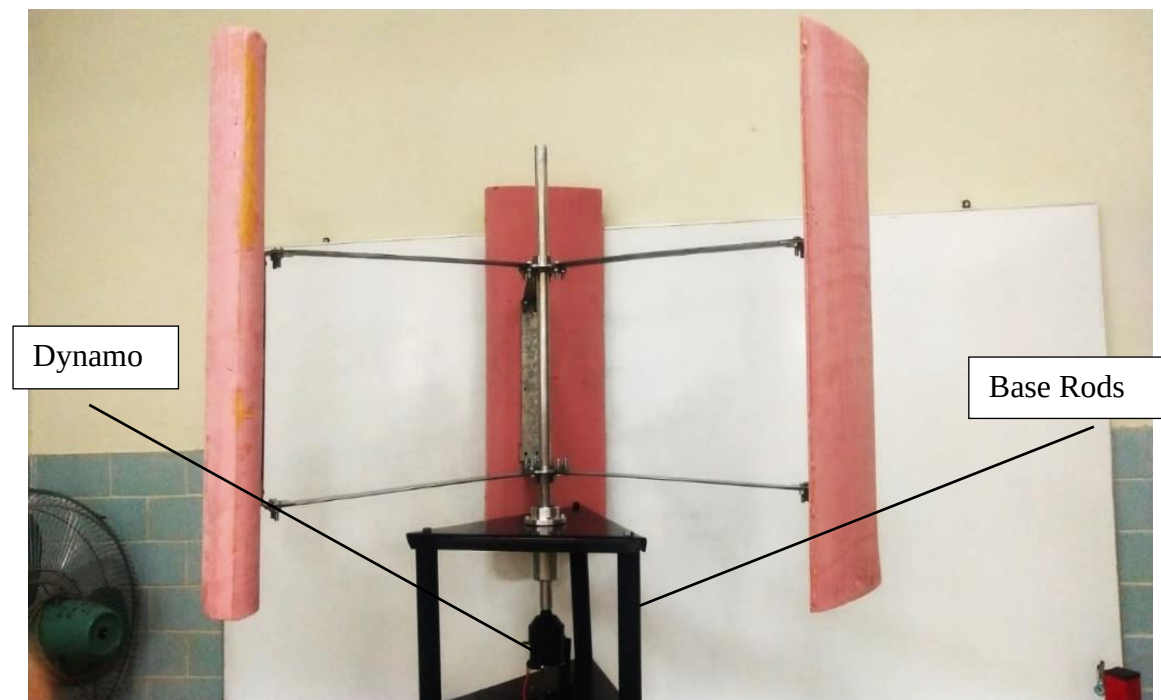


Fig 4-8: H-Darrieus Turbine

Assembly process of the H-Darrieus turbine began by connecting the struts to the retainers. After which shaft was inserted in the retainers. After connecting shaft to the struts, air foils are bolted to the struts via small wooden rectangular blocks. Once the rotor assembly has been completed, to connect the rotor with stator, shaft of the rotor has been pushed into the main retainer which is mounted on the base of the H-Darrieus turbine. Finally, the dynamo has been connected to the shaft to generate electricity. The H/D ratio remained same as of prototype that is 0.55. The following figure 4-8 shows the H-darrieus model assembly.

The following table details the components/parts utilized in the fabrication of H-Darrieus turbine.

Table 4-3: components/parts utilized in the fabrication of H-Darrieus turbine

Components/Parts	Quantity	Properties	
Aerofoil	3	Type	NACA0015
		Material	Jumbolon
Struts	6	Length	50 cm
		Thickness	mm
Retainers	2	Material	Mild steel
		Outer diameter	6.5cm
Bearings	1	Type	Radial Insert Ball Bearing UC 210
		Inner diameter	2.5 cm
		Outer diameter	4.2 cm
Dynamo	1	Input speed	Up to 9000rpm
		Max load Current	880mA
		Voltage	24V
Shaft	1	Length	80cm
		Diameter	2.5 cm
Plates (Base)	2	Material	Mild steel
		Shape	Triangular
Base rods	3	Material	Mild steel
		Diameter	6 cm
		Length	30 cm

4.7.2 Savonius Turbine

Savonius turbine provides ease of fabrication. It includes savonius blades, shaft, end plates, AC generator, top and base and three supporting rods to hold the structure of the rotor. The development process was initiated with the fabrication of the blades of the turbine. Dimensions were taken from the literature review and by considering the prototype results, PVC pipe having 10cm radius with 2mm thickness was selected. The development of blades began with the cutting of the pipe into three semi-circular pipes having length of 60cm. Iron strips having thickness 1mm were CNC milled into circular plates, with radius of 19cm, to form end plates. For fixing the blades, metal strips were bolted to the blades as shown in fig. These strips were welded with the end plates. To slip retainers into the end plates, a hole was cut at the centre of the end plates. For fixing the end plates with the retainer, screw holes of similar sizes were threaded into both, and nut bolt joint was made between retainer and end plates as shown in Fig. 4-9. The overlapping ratio for the blades

were kept being 20% and H/D ratio remained 3. Shaft was inserted into the retainers and to grip the shaft tighten, L key stud of 5mm was used in the retainers.

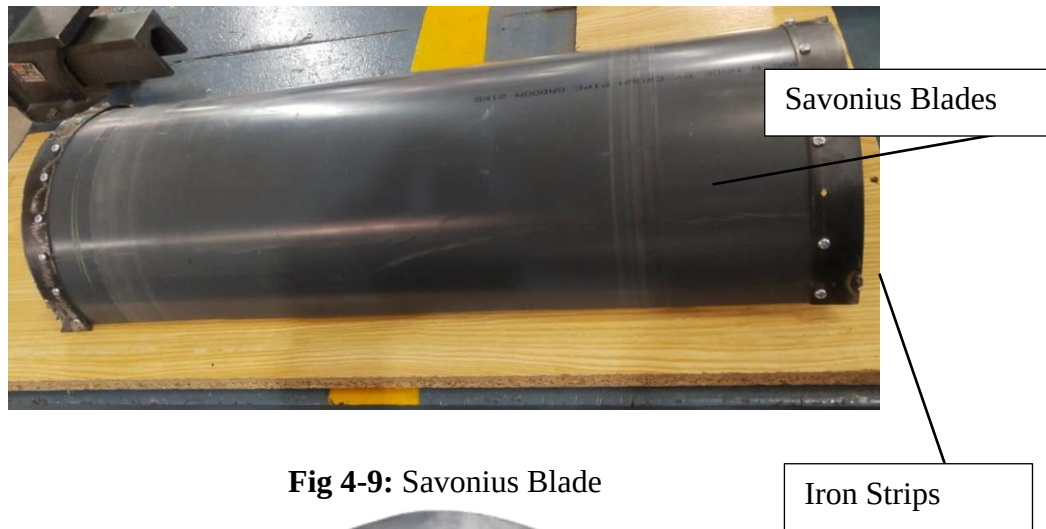


Fig 4-9: Savonius Blade

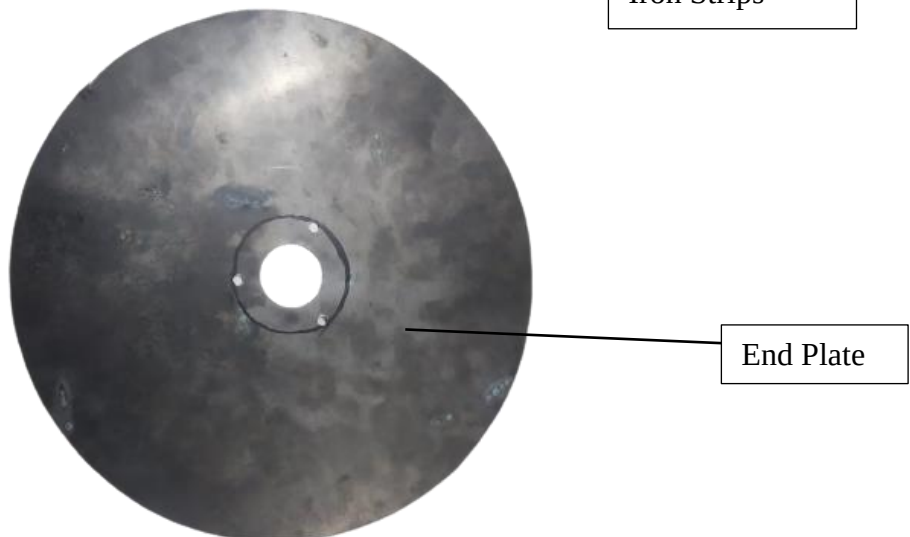


Fig 4-10: End Plate

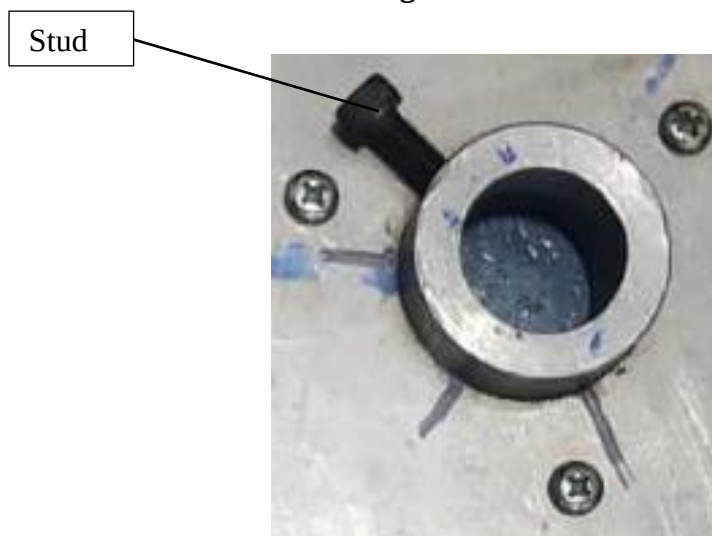


Fig 4-11: Retainer attached to end plate

Two radial insert ball bearings were used in the Savonius turbine. To secure the correct alignment, a fabricated bearing brackets (UC) were mounted on top and base using nut and bolt joint as shown in Fig.4-12. The top and base plates having bended edges were made of mild steel sheet. To hold the top plate and strengthen the structure of the turbine three vertical mild steel rods were bolted down. Three base (mild steel) rods were bolt jointed to support the base of the savonius turbine. Under the base plate, Dynamo was connected to the shaft to convert the rotational motion of the shaft into electrical energy.

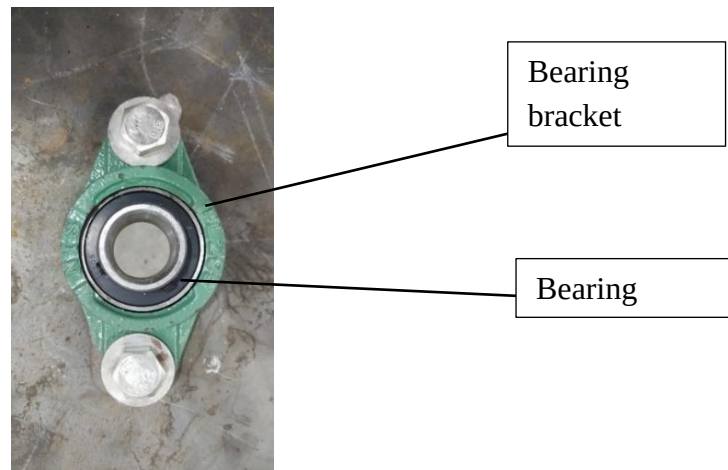


Fig 4-12: Bearing

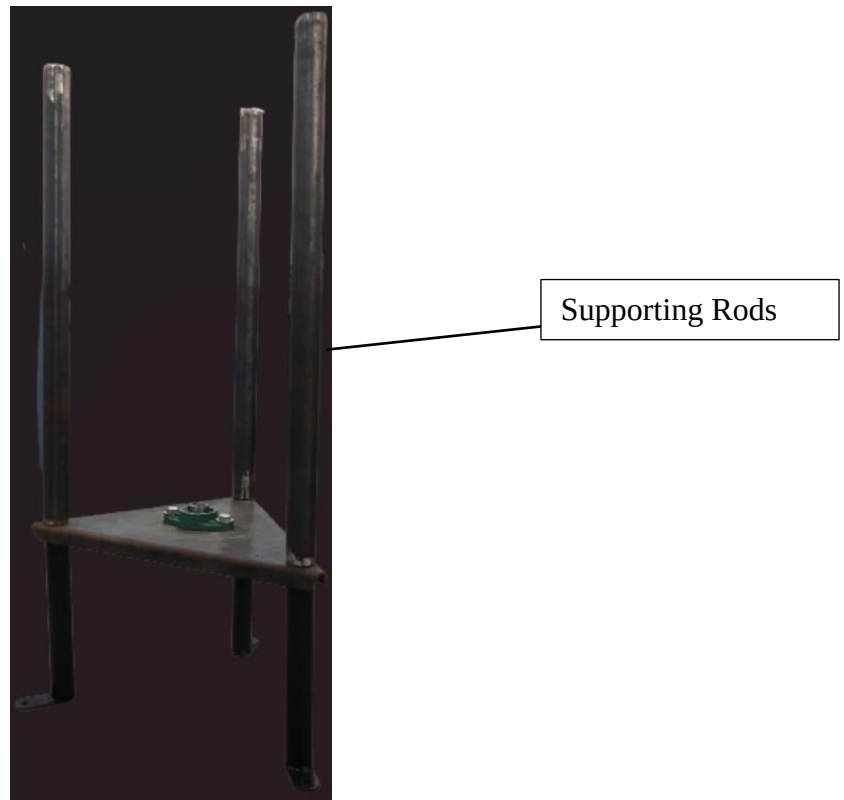


Fig 4-13: Supporting Structure of the Savonius turbine

Once the fabrication of the components of the savonius turbine has been completed, the assembly process of the savonius turbine has been initiated, by inserting the shaft in the retainers which were connected to the end plates. Blades were attached to the end plates through the welding of metal strips to end plates. Once the rotor has been constructed, shaft has been pushed through the bearing which was attached to the base plate and dynamo was connected to the shaft to produce electricity. Top plate was placed on the supporting vertical rod (which were connected to the base) such that shaft of the rotor slides through bearing which were mounted, via bearing bracket, on the upper side of the top plate.



Fig 4-14: Savonius Turbine

The following table presents the components/parts details which were utilized in the fabrication of savonius turbine.

Table 4-4: Components/parts details which were utilized in the fabrication of savonius turbine

Components/Parts	Quantity	Properties	
Pipes	3	Material	PVC
		Diameter	20cm
		Length	60cm
Shaft	1	Material	Aluminium
		Length	80cm
		Diameter	2.5cm
Endplates	3	Material	Mild steel
		Outer diameter	38cm
Supporting rods	3	Material	Pure Iron
		Length	70cm
Dynamo	1	Input speed	Up to 9000rpm
		Max load Current	880mA
		Voltage	24V
Bearings	2	Type	Radially Insert Ball Bearings UC210
		Inner diameter	2.5cm
		Outer diameter	4.5cm
plate (Base and top)	2	Material	Pure Iron
		Shape	Triangular
		Dimensions	2 sides of 30cm 1 side of 20cm
Base rods	3	Material	Pure Iron
		Diameter	2cm
		Length	25cm
Retainers	1	Material	Mild steel
		Outer diameter	4.5cm

4.7.3 Multi-stagging

Once the fabrication of H-darrieus and savonius turbine has been completed, multi-stagging process has begun. In multi-stagged turbine, savonius turbine has been placed on the ground while H-darrieus turbine had to be placed on top of it. Thus, to place H-darrieus turbine on the savonius turbine, three vertical supporting rods were used under the base plate of the

H-darrieus turbine such that H-darrieus turbine rests on the savonius turbine as shown in Fig. 4-15. The bottom end of these rods was connected to the upper end of the supporting rods of the savonius turbine using screw rods. The upper end of these supporting rods is bolted with base plate of the H-darrieus turbine.



Fig 4-15: Multi-Stage Vertical Axis Wind Turbine

4.8 Integration and instrumentation

The instrumentation of the Multi-stage VAWT focused on measuring the wind velocity, with which it strikes on the turbine blades, and turbine rpm using the tachometer. Anemometer was used to measure the wind velocity at the face of the turbine. The electrical equipment includes DC generator (dynamo) and Digital multi-meter. DC generator is connected to the shaft and converts the rotational energy of the shaft to electrical energy. The digital multi-meter measures the voltage produced by the DC generator. Two DC generator have been used, one on each shaft.

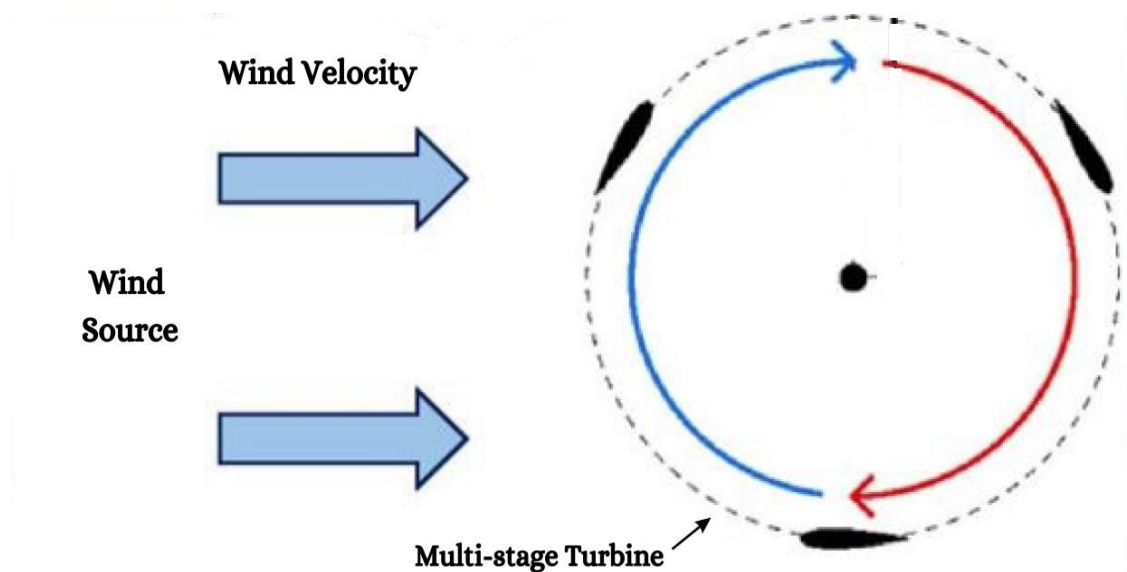


Fig 4-16: Experimental setup with a wind source

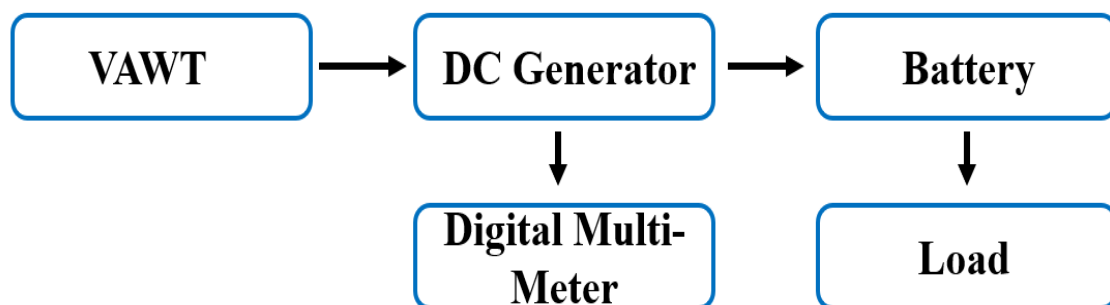


Fig 4-17: Schematic of Electronic Circuit

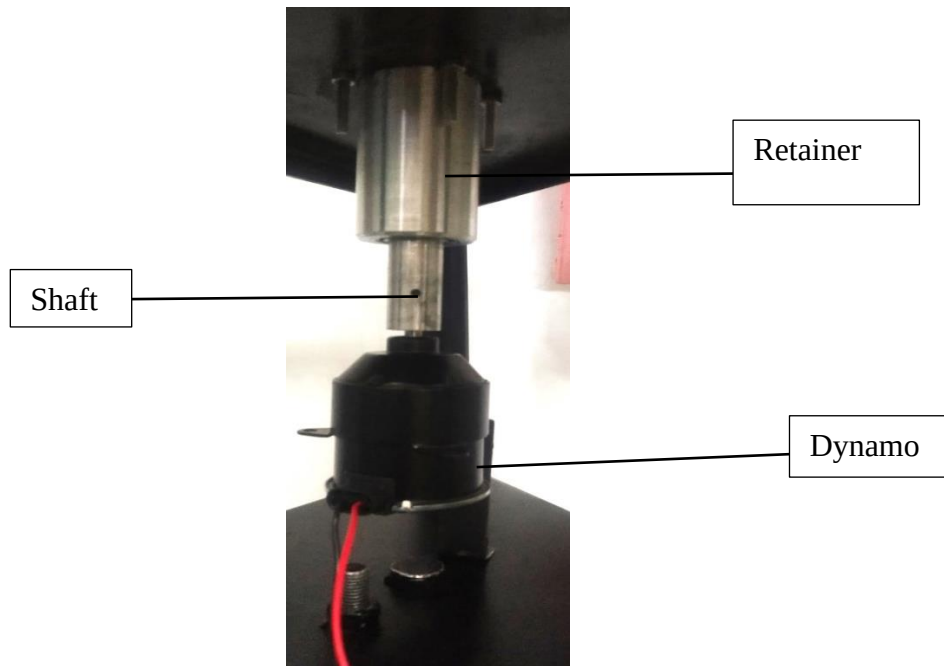


Fig 4-18: Power Generation System

4.9 Testing/Experimentation Procedures

The aim of testing and experimentation was to determine the feasibility of the multistage highway turbine. The experiment was carried out by directing the air flow towards the blades of the multi-stage turbine. The air source was placed on angle such that maximum amount of air produced by the wind source could be harnessed by the multi-stage turbine. Wind source provides the air velocity up to 10m/s. The velocity of the wind is measured using anemometer. Using anemometer, wind velocity produced is measured on different highways and the results are between 2m/s to 8m/s. Thus, the available wind source had the ability to provide the replica of the real-world scenario. Data was collected on various wind velocities, and various distances between wind source and the turbine. Readings for wind velocity, corresponding wind turbine rpm and voltage produced by the generator has also been noted down.

The following table is the form used to collect data for the wind turbine's input and output.

Table 4-5: form used to collect data for the wind turbine's input and output

SNO	Distance between fan 1 and turbine (m)	Distance between fan 2 and turbine (m)	Air velocity at wind turbine due to fan 1 (m/s)	Savonius rotor RPM	H-darrieus rotor RPM	Generated voltage (Savonius rotor) (V)	Generated voltage (H-darrieus rotor) (V)

4.10 Summary

This chapter covers the development process of prototype, full-scale model, integration and instrumentation, and testing procedure. The fabrication process of the prototype was done by extracting the design parameters from simulation. Prototype was testing in the wind tunnel. Obtained results ensured the feasibility of utilizing multi-stage turbine on highways. Thus, full scale model was developed, and DC generator had been installed on both shafts. Pictures of the components and assembled model has also been included in this chapter. Testing procedure for final model was simple as it only required a wind source, anemometer, tachometer, and digital multi-meter. Form to be utilized during the testing of the final model is also presented at the end of the chapter.

Chapter 5

RESULTS AND DISCUSSION

Testing of the turbine was done in wind tunnel which generates wind velocity and maximum we utilized is 8 m/s. Turbine was placed in wind tunnel at a range of wind velocities so that threshold of turbines and maximum rpm could be measured.

Parameters Needed to be Measured

- Fan rpm
- Wind Velocity
- Turbine rpm
- Torque Produced from Wind
- Voltage Generated
- Current Produced

For the testing of savonius turbine it was placed in wind tunnel and its speed was varied, and that speed was noted down, rotational speed of turbine was calculated using tachometer. For the proper working of tachometer, a reflector was attached to one of the wings of savonius turbine and tachometer measured the speed in rpm. To measure the wind velocity anemometer was used, fan of anemometer was placed near to the blades of turbine so that the velocity being experienced by blades would be measured.

5.1 Results

This portion discuss the results obtained from prototype of savonius and h-darrieus turbine. Testing was done in wind tunnel for both turbines separately and data was collected for the purpose of analysis and to decide what specifications would be needed to make full scale model. Other most important thing is to validate our simulations results with prototype results because validation of simulative results is a compulsory task to do. Prototype testing was done on three turbines h-darrieus, simple savonius and helical savonius. But blades of helical savonius were complex to make and weren't manufactured properly so results got from helical savonius were highly dissatisfying so proper testing and analysis was done of h-darrieus and simple savonius, theoretically h-darrieus is more efficient than simple savonius.

5.1.1 H-Darrieus

Table 5-1: Data for prototype of H-Darrieus

No. of Obs.	Wind velocity (m/s)	Turbine Angular velocity (rpm)	Voltage (V)	Current (A)	Power (W)	Torque (Nm)	Torque Coefficient (Prototype)	Torque Coefficient (Simulation)	Error (%)
1	3.2	68	0.66	0.4	0.267	0.003	0.05	0.06	20
2	4	110	2.3	0.8	1.84	0.018	0.157	0.18	14.64
3	5	170	6	1.292	6.56	0.038	0.2	0.22	10
4	6	208	10	1.589	15.89	0.077	0.28	0.31	10.71
5	7	259	12.3	2.8	33.9	0.132	0.35	0.4	14.2
6	8	304	14	4.9	69.4	0.189	0.42	0.45	7.142

5.1.1.1 Effect of Wind Velocity

It has been observed that wind velocity of 3.2 m/s is required to kick off the rotation of the rotor. As the wind velocity increases, turbine angular velocity increases which results in increase of torque and voltage generation by the turbine. The following graphs show the effect of wind velocity to the angular velocity of the turbine and torque generated by the rotor.

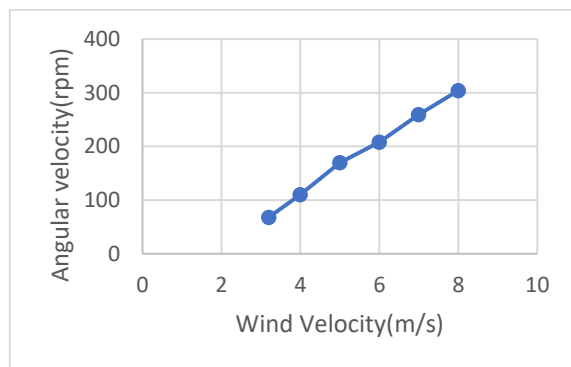


Fig 5-1: Turbine rpm vs Wind Velocity

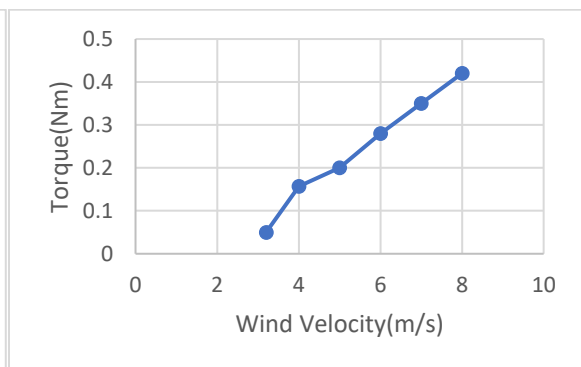


Fig 5-2: Torque vs Wind Velocity

5.1.1.2 Comparisons of Simulations and Prototype Results (H-Darrieus)

The trend in both graphs shows that turbine angular velocity and torque can be increased by increasing the wind velocity. When the blades start moving, torque in the turbine is

0.00393 Nm which increases up to 0.189 Nm as the wind velocity reaches 8m/s. The trend is almost for experimental results to simulation results.

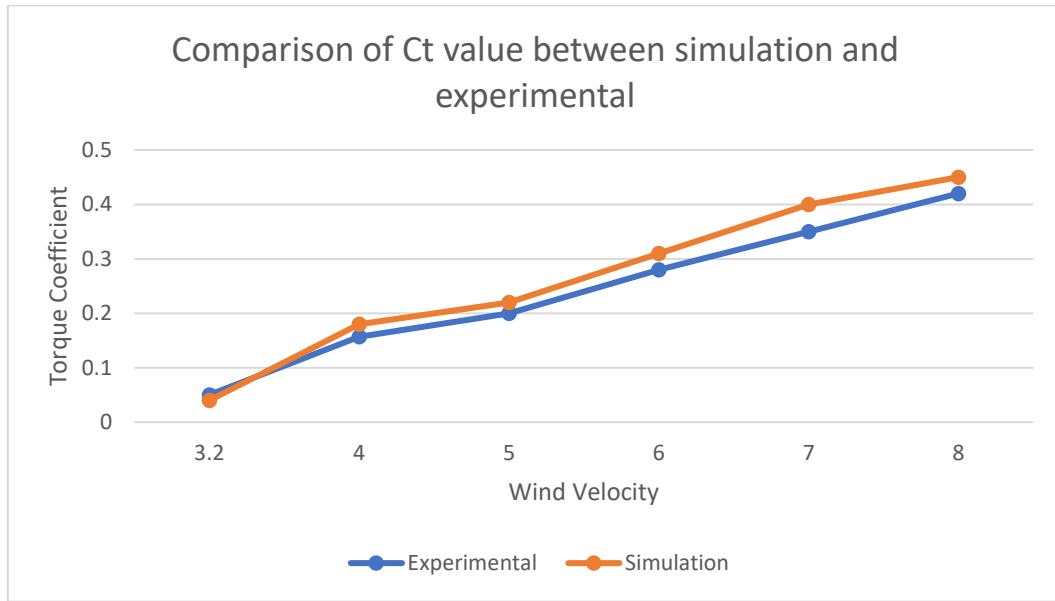


Fig 5-3: Comparison of Ct value between simulation and experimental

The above shown graph compares the torque coefficient value achieved by simulation and experimental setup. A little difference has been noticed in the values which is less than 20 percent. This difference occurs due to unfeasibility of ideal conditions for the experimental setup.

5.1.2 Savonius Turbine

Table 5-2: Data for Prototype of Savonius Turbine

No. of Obs.	Wind velocity (m/s)	Turbine Angular Velocity (rpm)	Voltage (V)	Current (A)	Power (W)	Torque (Nm)	Torque Coefficient	Torque Coefficient (Simulation)	Error (%)
1	2.9	70	0.35	0.00009	0.319	4.55×10^4	0.12	0.14	16.66667
2	4	100	0.62	0.00022	0.1417	1.4×10^3	0.21	0.23	9.52381
3	5	130	0.85	0.00035	0.303	2.31×10^3	0.22	0.24	9.090909
4	6	176	0.96	0.00071	0.69	3.9×10^3	0.26	0.28	7.692308
5	7	210	1.3	0.0008	1.062	5.5×10^3	0.27	0.29	7.407407
6	8	248	1.62	0.0011	1.87	7.5×10^3	0.28	0.31	7.1428

Savonius turbine requires less starting torque as compared to H-darrieus turbine. Thus, its blades start rotating at wind speed of 2.9 m/s. Turbine angular velocity, torque produced in the turbine and voltage generated by turbine are mainly dependent on the wind velocity. The following graphs show the behavior of turbine angular velocity and torque on varying wind velocity.

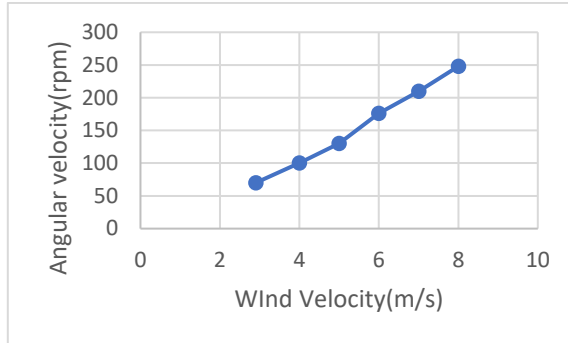


Fig 5-4: Turbine rpm vs Wind Velocity

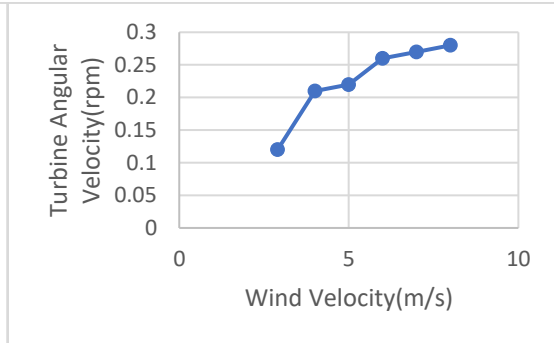


Fig 5-5: Torque vs Wind Velocity

The trend of these graphs is almost same as of the H-darrieus graphs. Initially, torque is 0.00045 Nm and reaches up to 0.0075Nm at wind speed of 8 m/s. The graphs clearly show that at the initial stage savonius turbine provides better results (high turbine angular velocity and higher torque) but as the wind velocity increases, H-darrieus becomes more prominent.

5.1.2.1 Comparisons of Simulations and Prototype Results (Savonius)

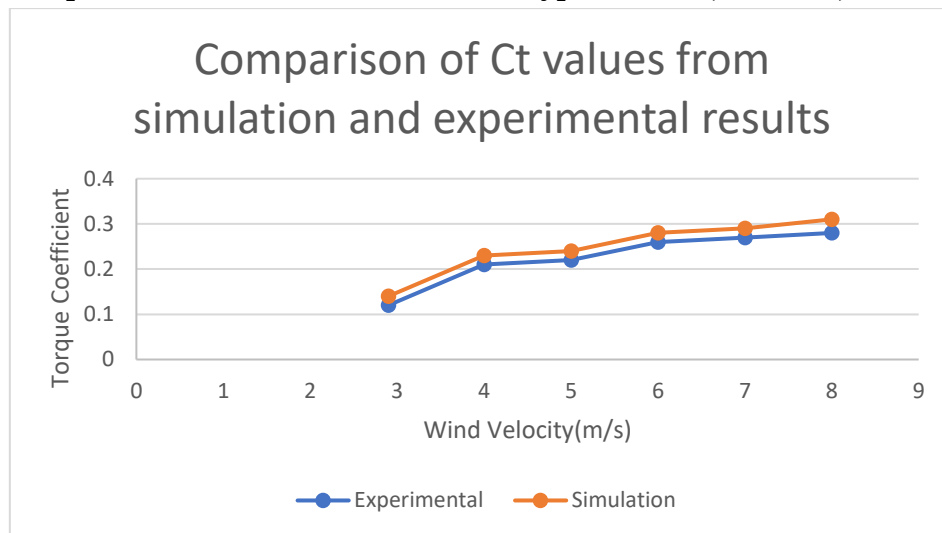


Fig 5-6: Comparison of Ct values from simulation and experimental results

A comparison of the results of experimental and simulation results have been shown in above Fig 5-6. It can be seen that for torque coefficient, little error occurs which has value of less than 20 percent. This error occurs due to the unfeasibility of the ideal conditions.

5.2 Full Scale Model

Results and analysis of testing of full-scale model starts from here. Full scale model is multi-staged wind turbine with savonius at the bottom and h-darrieus at top. Theoretically h-darrieus is more efficient as compared to savonius turbine.

5.2.1 Savonius Turbine

Testing of the turbine was done using pedestal fan which generates maximum wind velocity of 7m/s. Fan was placed at different distances from turbines to get a range of wind velocities so that threshold of turbines and maximum rpm could be measured.

Parameters Needed to be Measured

- Fan rpm
- Wind Velocity
- Turbine rpm
- Torque Produced from Wind

For the testing of savonius turbine fan was placed on ground level and its speed was varied by changing the position of fan and varying the speed of fan using the regulator. Data was noted down, rotational speed of turbine was calculated using tachometer. For the proper working of tachometer, a reflector was attached to one of the wings of savonius turbine and tachometer measured the speed in rpm. To measure the wind velocity anemometer was used, fan of anemometer was placed near to the blades of turbine so that the velocity being experienced by blades would be measured. The data observed is given below:

Table 5-3: Data for Savonius Turbine

S#	Distance from Fan (ft)	Fan rpm (rpm)	Wind Velocity (m/s)	Turbine rpm w (rpm)	Torque by Wind (Tw)
1	5.3	1350	2.1	71.67	0.06
2	5	1350	2.4	83.98	0.073
3	4	1009	3.1	95.62	0.127
4	3	1009	3.7	108.67	0.181
5	3.5	1350	4.2	120.54	0.233
6	2	1009	4.8	130.38	0.305
7	2.5	1350	5.3	138.87	0.373
8	2	1350	5.7	147.13	0.429
9	1	1184	6.2	153.61	0.508
10	1	1350	6.8	160.18	0.611
11	0.5	1350	7	169.27	0.647

From the data is concluded that threshold of savonius turbine is 2.1 m/s and rpm generated at that speed are 70 and maximum wind velocity generated by fan is 7 m/s and wind velocity at rpm of turbine at that speed are 169 rpm. Maximum torque generated by

the wind is 0.647 Nm at wind velocity of 7 m/s, this isn't the actual torque when actual torque will be calculated there will be some losses that are considered by coefficient of torque C_t but calculating that C_t requires value of power which comes from voltage and current and measuring those is beyond the scope of this FYP.

5.2.1.1 Effect of Wind Velocity on Turbine rpm

As shown in graph relationship between wind velocity and turbine rpm is linear, with increase in wind velocity turbine rpm also increases. Wind velocity varies from 1m/s to 7m/s and correspondingly turbine rpm varies between 70 rpm and 170 rpm. Maximum rpm is at wind velocity of 7 m/s. As shown in graph threshold wind velocity is 2.3 m/s and turbine rpm at that threshold is 70.

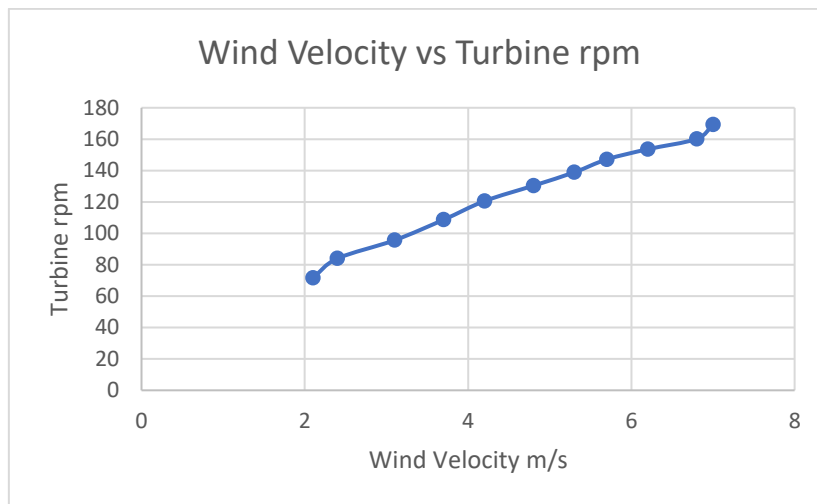


Fig 5-7: Graph between wind velocity and turbine rpm

5.2.1.2 Effect of Wind Velocity on Torque from Wind

Graph shows the linear relationship between torque and wind velocity as wind velocity increases torque also increases. Minimum torque generated from wind is 0.09 at wind velocity of 2.3 similarly at maximum wind velocity of 7 m/s values of torque is 0.67.

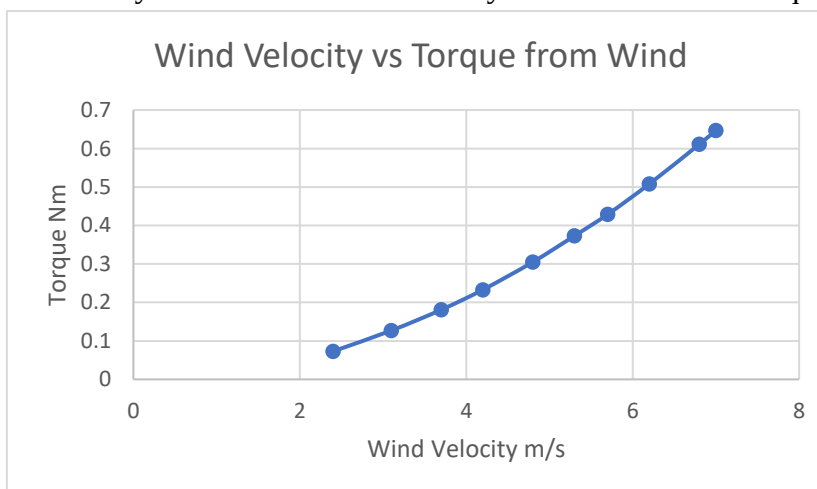


Fig 5-8: Graph between wind velocity and torque

5.2.1.3 Effect of Turbine rpm on Torque from Wind

As shown in graph relationship between torque from wind and turbine rpm is linear, with increase in turbine rpm, torque also increases. Turbine rpm varies from 70 to 170 and correspondingly torque varies between 0.05 and 0.65 Nm. Maximum torque is at turbine rpm of 170.

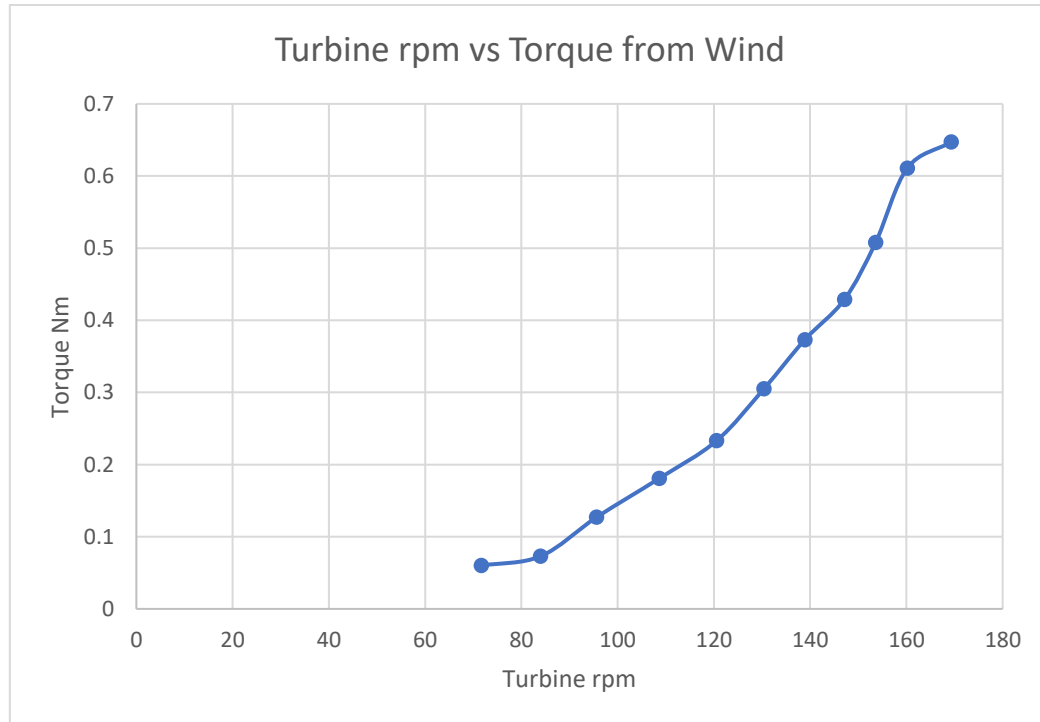


Fig 5-9: Graph between turbine rpm and torque

5.2.2 H-Darrieus Turbine

H-darrieus turbine works in a better way at some height from ground level so it was mounted on the top of savonius turbine. Fan's wind couldn't reach at that point, so fan had to be placed on the chair to make wind velocity approach the airfoils of turbine. Same phenomenon was followed in the testing of h-darrieus as was in savonius. Wind velocity was varied by changing the distance between fan and turbine and by altering the fan rpm using regulator.

Parameters Needed to be Measured

- Wind Velocity
- Turbine rpm
- Fan rpm
- Torque Produced by Wind

Wind velocity was measured using anemometer by placing it near to the airfoils of turbine so that wind velocity experienced by them would be considered. To measure the rpm of turbine a reflector was attached at the bottom of an airfoil and tachometer was used to calculate the rpm. Maximum and minimum rpm were found out by placing the fan as close as possible to the airfoils. Wind velocity at which rpm was found to be minimum was declared threshold of h-darrieus. Data measured is given below:

Table 5-4: Data for H-Darrieus Turbine

S#	Distance from Fan (ft)	Fan rpm (rpm)	Wind Velocity (m/s)	Turbine rpm w (rpm)	Torque by Wind T_w (Nm)
1	5	1009	3	35.65	2
2	5.3	1350	3.1	37.71	2.1
3	4.8	1350	3.2	40.45	2.276
4	4	1009	3.3	45.19	2.421
5	3	1009	3.5	51.67	2.723
6	3.5	1350	4.1	60.06	3.738
7	2	1009	4.8	72.38	5.12
8	2.5	1350	5.3	85.87	6.247
9	2	1350	5.7	91.13	7.224
10	1	1184	6.2	100.61	8.532
11	1	1350	6.8	111.18	10.282

After analyzing the data maximum rpm was found out to be 115 at 7 m/s on the other hand threshold wind velocity is between 3 to 3.3 m/s and rpm at that threshold are 35 to 45. Maximum torque generated by the wind is 10.282 Nm at wind velocity of 6.8 m/s, this isn't the actual torque when actual torque will be calculated there will be some losses that are considered by coefficient of torque C_t but calculating that C_t requires value of

5.2.2.1 Effect of Wind Velocity on Turbine rpm

As shown in graph relationship between wind velocity and turbine rpm is linear, with increase in wind velocity turbine rpm also increases. Wind velocity varies from 1m/s to 7m/s and correspondingly turbine rpm varies between 30 rpm and 118 rpm. Maximum rpm is at wind velocity of 7 m/s. As shown in graph threshold wind velocity is 2.9 m/s and turbine rpm at that threshold is 30.

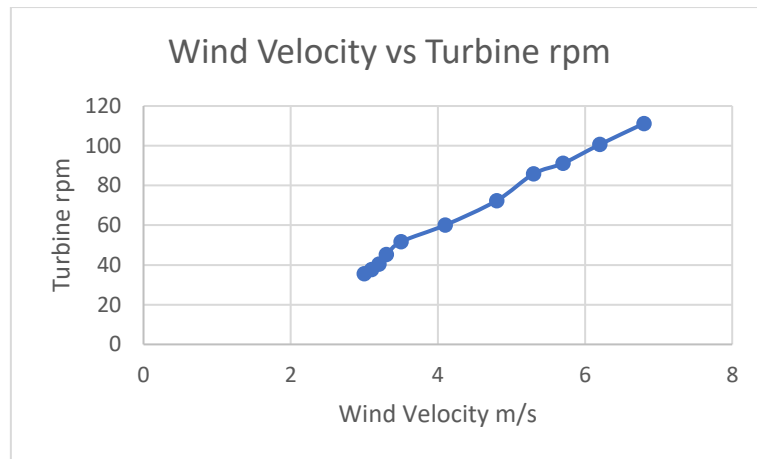


Fig 5-10: Graph between wind velocity and turbine rpm

5.2.2.2 Effect of Wind Velocity on Torque from Wind

Graph shows the linear relationship between torque and wind velocity as wind velocity increases torque also increases. Minimum torque generated from wind is 2 Nm at wind velocity of 3.3 similarly at maximum wind velocity of 7 m/s values of torque is 10.68 Nm.

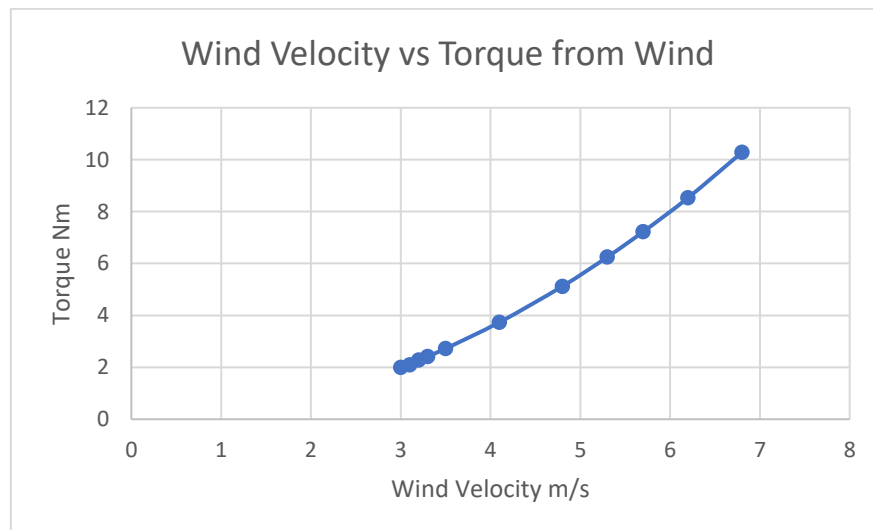


Fig 5-7: Graph between wind velocity and torque

5.2.2.3 Effect of Turbine rpm on Torque from Wind

As shown in graph relationship between torque from wind and turbine rpm is linear, with increase in turbine rpm, torque also increases. Turbine rpm varies from 30 to 110 and correspondingly torque varies between 2 and 10.68 Nm. Maximum torque is at turbine rpm of 110.

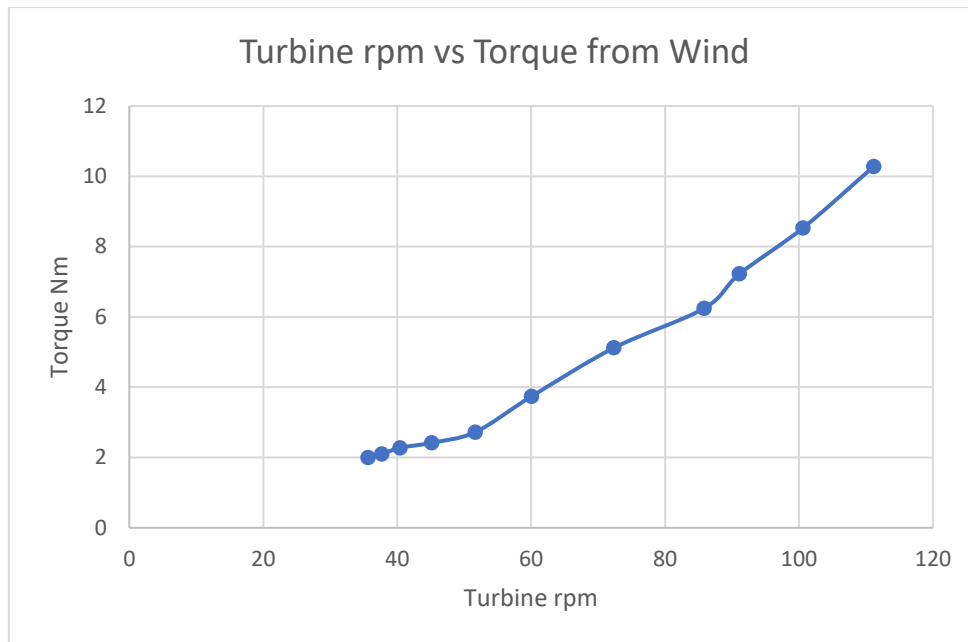


Fig 5-7: Graph between turbine rpm and torque

5.3 Analysis and Discussion

If the comparison of these results is done with the simulated results it can be concluded that simulation was right and simulated data is validated by experimental observation. Threshold wind velocity for Savonius Turbine according to simulations was 2.5 m/s, experimental observation shows threshold is between 2 and 2.3 m/s. Data is within 15% of error for Savonius Turbine. Comparing the simulated data of H-Darrieus with experimental one it can be seen that in simulated data threshold wind velocity is 2.5 m/s and experimentally it was concluded that threshold of H-Darrieus turbine is between 3 and 3.5 m/s. So, simulations of both turbines are validated with experimental data.

5.4 Summary

This chapter targets the detailed discussion of results and analysis of prototype and full-scale model. Data is collected from experimentation and simulation of CAD model after that both the results are compared, and it is concluded that simulative results are valid with the percentage error whose maximum value is 20%. Results of prototype and full-scale model couldn't be compared because of the unavailability of proper dynamo for full scale model because results needed to be validated by measuring voltage and current to go for power, so comparison of prototype and full-scale model isn't done.

Chapter 6

IMPACT AND ECONOMIC ANALYSIS

The increased usage of wind energy in the power production sector has resulted in significant advancements in wind turbine technology in recent years. Windmills have evolved into Horizontal Axis wind turbines (HAWT), and horizontal axis wind turbines have given way to Vertical Axis wind turbine (VAWT) designs. With the various evolutions that wind turbines have undergone, the impact they have had at various eras has also changed. With the change in various designs, it also impacts the environment in different ways due to its structure, size, and sustainability.

6.1 Social Impact

Energy obtained from the clean and green source has the huge impact on society. Near the powerplants and conventional wind farms, people have an issue of noise, vibration, and pollution in case of powerplant. Noise and vibrations were observed by adjacent populations as a result of the installation of horizontal axis wind turbines. We have greatly lowered noise levels with the VAWT, and there are much lower magnitude vibrations with the lower structure and harvesting is better than the previous model. Pollution can also be reduced in the areas where the VAWT farms are installed and due to smaller size, a land will be available for trees and other purposes.

Due to sharp blades of HAWT, it kills the birds in its surrounding. But due to compact structure it does not affect the birds. We decreased noise and the possible risks associated with it by using the VAWT compact design. The compact design also helps to reduce the amount of room required for a Savonius turbine. These, unlike HAWT, do not require a lot of area to operate and may be placed near together. These clever developments have contributed to the preservation of lands that can be used for other purposes. There will be significant short-term employment creation during project construction, as well as more elusive job generation throughout project operation and maintenance.

6.2 Sustainability Analysis

Wind energy is widely acknowledged as an important component of the global effort to reduce greenhouse gas emissions. Because the generation is depending on the wind speed in the area, we efficiently eliminate any fuel costs and utilize the air speed for energy generation. With the development of the Savonius and H- Darrieus wind turbine multi-staging, we are taking another step toward more sustainable and cleaner electricity generation and get efficient results than the conventional single stage VAWT. We are reducing our reliance on traditional power generation technologies, and this is the way power generation will evolve in the future. Renewable energy generation has a bright future since it can be utilised over and over again with little to no negative environmental impact.

These resources can help mitigate the greenhouse effect and global warming effect because they are considered clean energy resources.

The installation of different turbines can be a very tricky point but as the Savonius wind turbines are drag type turbines, which means they do not need to be positioned at a height, allowing them to be installed below and harvest the wind energy created by moving vehicle and as the H-Darrieus is the lift type and need high speed of air to generate moment so, installed on the upper portion to capture maximum energy. As a result, the issue of installation at a height or in a windy place is not an issue like the HAWT; however, areas with a wind speed of 3 m/s or higher are suitable for VAWT to function in, and they can operate in most weather situations. These VAWT may be built close together, occupying less space, and tests have shown that a combination of both can capture more energy than the single one.

It has also been tested that the farm of VAWT can capture more energy than the conventional HAWT farm with much less cost. The manufactured model can start to capture energy from 3 m/s to 50 m/s , which is the speed of air at most of farms. The model can be installed at the coastal areas as well as the barriers between the roads so, to harvest energy by moving vehicles. The structure model is made from mild steel and aluminum, blades of savonius is made from PVC pipes and air foils are made from jumbolon to reduce the weight of the model and make more efficient. Blades of savonius can also be made from carbon fiber to reduce weight and cost. These materials use in the project can be readily available in the market without any restrictions and limitations.

Another factor must be considered is the maintenance of the VAWT, it does not require much maintenance once the generator that is installed below the base is covered with the box. It can operate in any weather conditions. During the thunderstorm, the harvesting capacity of the VAWT increased many times depend on the air speed.

6.3 Environmental Impact

The process of generating electricity by the combination of VAWT is the clean and green process and it helps us to utilize energy without any carbon emission. So, moving towards these methods can help us to reduce the greenhouse effect on the environment and will help to live in a better world. By harvesting the energy using this method, there is no waste generated. The dust on the blades can be the waste which can be washed by the water and can be duped in nearby fields which is non-hazardous to any plant and animal.

The development of the production process included CNC milling of mild steel and aluminum work pieces, extrusion of the PVC pipe and manufacturing of jumbolon board needed for the blades. The sole course of action that contributes to the greenhouse effect is to operate this equipment. Aside from that, the project is far more eco-friendly. If this equipment can shift on the energy that is made from clean source, it can be made completely eco-friendly. Greenhouse gas emissions are also reduced, which is good for the health of the people who live nearby.

6.4 Sustainable Development Goals (SDGs)

Table 6-1: Table showing sustainable development goals

S. No	SDG	Adherence of FYP to SDG
1.	Affordable and Clean Energy	The suggested design offers the community with economical, clean, and green energy
2.	Decent Work and Economic Growth	Installation on Vertical Axis Wind Turbine (VAWT) on a large scale will help in the economy growth of the country
3.	Industry, Innovation, and Infrastructure	The large-scale implementation of a concept will transform the industry and aid in the development of new infrastructure
4.	Partnerships	Support funding from the local government or any other firm can assist in resolving energy shortages on the highways
5.	Life on Land	Birds will be safe from VAWT as compared to HAWT
6.	Good Health and Well-being	VAWT will minimize the energy shortage and will be for well-being of people and it has harmful impact
7.	Responsible Consumption and Production	N/A
8.	Gender Equality	N/A
9.	Clean Water and Sanitation	N/A
10.	Zero Hunger	N/A
11.	Reduced Inequality	N/A
12.	Quality Education	N/A
13.	Sustainable Cities and Communities	N/A
14.	Climate Action	N/A
15.	Life Below Water	N/A
16.	No Poverty	N/A
17.	Peace and Justice Strong Institutions	N/A

6.5 Hazards Identification and Safety Measures

As the structure of the VAWT is smaller than any other turbine, it cannot withstand high speed of air and the wind may carry the entire model, thus a temporary joint with the model's base rods and the ground must be made to keep the structure in place.

When water gets into touch with the winding of the generator, it generates a tremendous loss and may cause the entire system to become short during wet weather or rainy conditions. To avoid a situation like this, a generator should be kept in a box with safety precautions, and each VAWT should have a first-aid kit to cope with any eventuality.

6.6 Summary

In this chapter, social impact analysis, sustainability analysis, environmental impact analysis and the hazardous related to the proposed design of VAWT are discussed and the safety precautions we must have taken to reduce these hazards are also discussed. Renewable energy is viewed as a distinct power source from traditional fossil fuels. Because of the depletion of fossil fuels and its impact on climate change, renewable energy is becoming increasingly important. The project's major goal was to be sustainable. The blades of a wind turbine (vertical or horizontal) rotate around the rotor, which allows the generator to create electricity, converting wind energy to electricity. The final proposed design has minimal social and environmental impact which is about to none, and it is one of the most sustainable designs develop till date. Other than that, the project is also cost effective.

Chapter 7

CONCLUSION

Renewable Energy is considered to be the one of the most reliable sources to fulfil the future requirements. It is getting more common due to unlimited availability of renewable resources while giving no harm to the environment. Renewable energy provides a sustainable source of the energy which is the main motive behind the project. Wind turbines produce electricity by converting the rotational kinetic energy into electrical energy. A design for wind turbine, to be utilized on highways to convert wind energy produced by the moving vehicles to electrical energy, was proposed in the form of multi staged Vertical Axis Wind Turbine (VAWT). VAWT was preferred over horizontal axis wind turbine due to their low starting torque and having almost no sound. Three most efficient designs were taken from the literature review and CFD analysis have been on ANSYS. After analysing CFD results, it was decided to perform the multi-stagging H-darrieus and savonius turbine to get the highest achievable efficiency. Savonius turbine utilizes the drag force to rotate while H-darrieus is lift base device to multi-stagging. VAWT Prototype has been fabricated to validate the simulation results. Testing was done by placing the prototype inside the wind tunnel. Obtained results were having less than 10 percent variations as compared to simulation results, which ensures the feasibility of utilization of multi-stage turbine on highways. After success of prototype, next step was to fabricate the full-scale model which could be utilized on highways. Dimension for the final model were taken from the literature review while keeping height of different types of vehicles in mind. Fabrication of full-scale model was initiated with the development of H-darrieus turbine after which savonius turbine was made. Finally, both of them are assembled. Once the fabrication has been completed, testing was performed by placing a wind source in front of the turbine, Wind Velocity was varied via wind source, and the corresponding rpm has been measured. Torque was calculated by utilizing rpm values. Using DMM values for voltage produced by the turbine were also noted down. The experimental data confirmed that, savonius turbine has the lowest starting torque while H-darrieus provides a higher efficiency at higher wind speeds. This further proves the importance of multi-stagging as it covers both cases of heavy traffics and low traffic. As savonius turbine is placed at bottom, it will work with the air produced by the cars, while H-darrieus will be moved by a bus or other heavy vehicles. Thus, this turbine is one of the most feasible sources of utilizing the wind energy produced by the moving cars and to produce the unlimited power for the highway lights and other public amenities.

7.1 Future Recommendations

This study is mainly focused on the mechanical design aspects of the turbine. Wind turbine with the most suitable mechanical design for highway use has been proposed but there are

several details which were not focused on this research. Though the mechanical efficiency has been significantly improved but it was outside the scope of the research to do the detail analysis of electrical components. Thus, designing the electricity generator which would convert the maximum amount of rotational energy of the rotor to electrical energy would be the great topic for further studies. For this purpose, the construction of induction motor would be suggested. Windings over metallic conductor, having cylindrical shape, and neodymium magnets placed in the slots will make the induction motor more suitable for this turbine. Furthermore, selection of other electronic components like battery etc could also play a cameo role in increasing the efficiency of the turbine.

Another suggestion to improve the efficiency of the developed multi-staged vertical axis wind turbine is to improve the design of H-darrieus by utilizing the space between the struts by placing a thin sheet having low weight. As it will create more drag which will result in increase in rpm of the turbine, Moreover, utilization of gears in the turbine will also make it more efficient as it will increase the rpm of shaft of the generator.

APPENDIX

NACA 0015							
$\bar{U}$	c/D	H/D	σ	AR	H	D	c
$A = 1 \text{ m}^2$							
3	0.170	0.45	0.51	2.6	0.671	1.491	0.253
4	0.155	0.50	0.47	3.2	0.707	1.414	0.219
5	0.155	0.60	0.47	3.9	0.775	1.291	0.200
6	0.160	0.60	0.48	3.8	0.775	1.291	0.207
7	0.170	0.60	0.51	3.5	0.775	1.291	0.219
8	0.170	0.55	0.51	3.2	0.742	1.348	0.229
$A = 4 \text{ m}^2$							
3	0.110	0.50	0.33	4.5	1.414	2.828	0.311
4	0.095	0.55	0.29	5.8	1.483	2.697	0.256
5	0.090	0.60	0.27	6.7	1.549	2.582	0.232
6	0.065	0.45	0.20	6.9	1.342	2.981	0.194
7	0.065	0.45	0.20	6.9	1.342	2.981	0.194
8	0.065	0.40	0.20	6.2	1.265	3.162	0.206
$A = 9 \text{ m}^2$							
3	0.095	0.60	0.29	6.3	2.324	3.873	0.368
4	0.080	0.65	0.24	8.1	2.419	3.721	0.298
5	0.070	0.70	0.21	10.0	2.510	3.586	0.251
6	0.065	0.85	0.20	13.1	2.766	3.254	0.212
7	0.065	0.80	0.20	12.3	2.683	3.354	0.218
8	0.065	0.70	0.20	10.8	2.510	3.586	0.233

Fig 4-41: Appendix

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